

FACULTY OF COMPUTATIONAL AND SOFTWARE ENGINEERING

Elite Ethiopia, The digital Service and Marketplace Hub

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Declaration

We hereby declare that this final year project titled **"Elite Ethiopia System"** is the original work of the undersigned. All the ideas, information, analyses, designs, and implementations presented in this project are authentic and have been carried out solely for academic purposes. This project has not been submitted, either wholly or in part, for the award of any degree or diploma at any other institution.

All sources of information, whether primary or secondary, used in the preparation of this project have been properly acknowledged and referenced. We are fully aware of the academic regulations concerning plagiarism and academic dishonesty and affirm that this work complies with the ethical standards and guidelines set by the institution.

We take full responsibility for the originality and authenticity of this project and confirm our commitment to maintaining the highest standards of academic integrity.

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Abstract

The increasing adoption of digital technologies in Ethiopia has created new opportunities for online business, freelancing, and digital content creation. However, Ethiopian professionals and creators face significant challenges in accessing global platforms due to currency restrictions, limited payment integration, and regional accessibility barriers. As a result, many skilled individuals, freelancers, and digital product creators lack a suitable online environment that supports local transactions and aligns with the Ethiopian economic and technological context.

Elite Ethiopia is a unified digital platform designed to address these challenges by integrating a full-service freelancing system with a digital product marketplace. The platform enables users to offer or request professional services, sell or purchase digital content such as ebooks, music, videos, software, and graphic designs, and collaborate through real-time messaging. It incorporates secure local payment gateways, user profile management, ratings and reviews, and administrative monitoring tools. The system aims to empower Ethiopian freelancers, businesses, and creators by providing an accessible, reliable, and localized digital environment.

This document outlines the functional and non-functional requirements of the platform. It provides a comprehensive description of the system architecture, interfaces, user roles, feature modules, performance expectations, data structures, and security considerations. The document serves as a reference for developers, analysts, supervisors, and stakeholders to ensure consistent understanding and effective implementation of the system.

List of Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
DB	Database
ERD	Entity Relationship Diagram
EE	Elite Ethiopia
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SRS	Software Requirements Specification
UAT	User Acceptance Testing
UI	User Interface
UML	Unified Modeling Language
VPS	Virtual Private Server

Chapter One: Introduction

1.1 Introduction

The rapid growth of digital technologies across the world has transformed how people work, communicate, and exchange digital value. In Ethiopia, this transformation is steadily increasing as more individuals gain access to smartphones, the internet, and digital financial systems. Despite this growth, there are still limited online platforms tailored to the Ethiopian context—platforms that allow freelancers to offer services, individuals to hire skilled professionals, and creators to sell digital products securely using local payment systems.

Global freelancing platforms such as Fiverr, Upwork, and Freelancer.com offer opportunities for remote work but pose significant accessibility challenges for Ethiopian users. These challenges include restrictions on account creation, inability to withdraw income through local banks, and currency limitations such as the inability to use PayPal or international cards. As a result, many Ethiopian youths, freelancers, and talented creators remain excluded from the global digital economy.

To address these challenges, the **Elite Ethiopia** website is designed as a fully localized, accessible, and secure online platform that connects freelancers and digital creators with clients and businesses. The system provides an environment where users can create service offerings, buy or sell digital products, communicate through real-time messaging, and make secure payments using Telebirr.

The platform aims to stimulate digital entrepreneurship, empower skilled individuals, and enable organizations to access services without geographical limitations. By serving as both a freelancing marketplace and a digital product store, Elite Ethiopia represents an innovative approach to bridging the digital divide and expanding economic opportunities within Ethiopia.

1.2 Background Information of the Organization

The Faculty of Computing and Software Engineering (FCSE) is part of the Arba Minch Institute of Technology (AMiT), which has a rich history dating back to 1986 when it was initially established as the Arba Minch Water Technology Institute (AWTI). The institute aimed to address

Ethiopia's need for skilled professionals in water resource management. Over the years, it expanded its programs to include multiple engineering disciplines, including computer science and software engineering. In 2011, the institute was restructured under the Engineering Capacity Building Program (ECBP) to create AMiT, an autonomous institution dedicated to producing skilled engineers and promoting technology transfer. Today, FCSE is one of the five key faculties within AMiT, offering various undergraduate and postgraduate programs in computing and software engineering.

1.2.1 Vision of FCSE

To become a center of excellence in computing and software engineering education and research in Africa.

1.2.2 Mission of FCSE

To provide high-quality education, research, and community services in the fields of computing and software engineering. To produce competent graduates who can contribute to technological advancements and solve real-world problems.

1.3 Background of the Project

Over the past decade, Ethiopia has seen a significant expansion in digital literacy, online service adoption, and the use of local digital payment systems. The introduction of platforms such as Telebirr, , Commercial Bank of Ethiopia’s , and various fintech innovations indicates strong momentum toward digital transformation. However, despite improvements in digital financial infrastructure, Ethiopia still lacks a robust online ecosystem that supports freelancers and digital creators.

A large portion of Ethiopian youth have skillsets in software development, graphic design, content creation, music production, marketing, and other digital fields. Yet opportunities for monetizing these skills remain limited. Many rely on social media, informal groups, or manual communication channels to find clients—methods that are inefficient, unscalable, and prone to misunderstandings or fraud.

Parallel to this, Ethiopian businesses often struggle to find reliable service providers. There is no centralized online platform where qualified freelancers are categorized, rated, and easily accessible. This gap creates inefficiencies in the local digital services market.

The background of this project is therefore rooted in addressing several socio-economic and technological gaps:

- The need for a **localized freelancing platform** that works with Ethiopian payment systems.
- The lack of an **organized digital product marketplace** for selling software, ebooks, graphics, and digital media.
- The absence of a **trusted communication and rating system** to promote transparency between clients and freelancers.
- The growing digital economy in Ethiopia that requires platforms aligned with local needs and regulations.
- The academic requirement for a meaningful final-year project with real-world relevance and future market potential.

Elite Ethiopia emerges as a solution designed to empower Ethiopian digital workers, support local businesses, and contribute to national digital transformation goals.

1.4 Team Composition

The project is developed by a team of final-year Software Engineering students. Each member has a specific role to ensure the successful completion of the system. A typical composition includes:

Table 1: Team Composition

No	Name	ID.No	Email/Phone	Responsibility
1	Nancy Tesfaye	NSR/1221/14	969870920	All Activities
2	Serkalem Delelegn	NSR/1967/14	962042225	All Activities

3	Abel Mekonnen	NSR/2122/14	987371272	All Activities
4	Elsa Hussen	NSR/852/14	991132272	All Activities
Date	May , 2026			
Advisor	Mrs. Anteneh		941988731	

Task and schedule of the project

Activity	Dec ('25)	Jan	Feb	Mar	Apr	May
Project Proposal						
Requirement Analysis						
System Design						
Implementation & Coding						
Testing and Project Defence						

Table 2: Task and schedule of the project

1.5 Statement of the Problem

Ethiopian freelancers and digital creators face numerous obstacles that limit their participation in online digital markets, including:

- **Inaccessibility of global freelancing platforms** due to currency restrictions
- **Lack of support for local payment systems**
- **No centralized digital marketplace** for selling Ethiopian-developed digital products
- **Difficulty in finding clients or reliable service providers**
- **Absence of rating and review systems**, causing trust and quality issues
- **Manual communication channels** that lead to miscommunication and disputes

- **No structured dispute resolution mechanisms**
- **Lack of opportunities for youth** who have digital skills but limited platforms to use them

These challenges result in:

- Lost income opportunities
- Underutilization of talent
- Inefficient service delivery
- Reduced business productivity
- Limited digital economy participation

Therefore, a centralized, locally supported platform is required to solve these issues and provide a structured digital marketplace.

1.6 Objective of the Project

1.6.1 General Objective

To design and develop a secure, scalable, and user-friendly web platform that supports freelancing services and digital product transactions for Ethiopian users.

1.6.2 Specific Objectives

The specific objectives of this project are:

- To analyze the existing freelancing and digital product selling practices in Ethiopia
- To identify problems and limitations of the existing system
- To gather and document functional and non-functional requirements of the proposed system
- To design system models including use case diagrams, sequence diagrams, activity diagrams, and class diagrams
- To select appropriate frontend, backend, and database technologies for system development
- To design user interface prototypes for the proposed system
- To implement core modules of the system based on the approved design

- To test the system for correctness, usability, performance, and security
- To document the system and prepare it for deployment and future enhancement

1.7 Feasibility Analysis

1.7.1 Operational Feasibility

The Elite Ethiopia project aims to modernize and simplify the way freelancers and digital creators connect with clients across Ethiopia. Currently, many freelancers rely on informal communication channels such as social media groups or manual messaging, which lack structure, transparency, and security. Similarly, clients struggle to find reliable service providers or trustworthy platforms to purchase digital products.

The proposed system introduces a centralized platform where users can offer and request services, upload digital products, communicate in real time, and complete payments through local payment gateways such as **Telebirr,** , and . The system will be user-friendly and accessible through any device with a web browser (mobile, tablet, and desktop).

The platform integrates smoothly with the workflows of freelancers and businesses by offering core tools such as:

- Service posting and browsing
- Digital product uploading
- Real-time chat for negotiation
- Ratings and reviews
- Local payment processing
- Order tracking and dispute handling

Administrative tools (user management, service moderation, analytics) ensure that platform operations remain transparent and efficient.

Possible operational challenges include initial unfamiliarity with the system, hesitation to adopt a new digital marketplace, or difficulty in uploading large digital products for the first time. These

challenges will be addressed through user tutorials, onboarding guidance, and clear documentation.

Once implemented, the system is expected to:

- Reduce communication barriers
- Increase visibility for freelancers
- Improve trust through ratings and dispute resolution
- Enhance transparency in transactions
- Promote efficient digital commerce
- Expand income opportunities for users across Ethiopia

1.7.2 Technical Feasibility

The project uses modern, reliable, and widely available web technologies that are appropriate for building scalable public websites.

- *Technology Availability*

- **Frontend:** React.js, NextJS
- **Backend:** Node.js / Express.js
- **Database:** MySQL
- **Real-Time Messaging:** WebSockets (Socket.io)
- **Payment Integration:** Telebirr API
- **Hosting:** VPS compatible with the system
- **Version Control:** Git & GitHub
- **Deployment:** cPanel

These technologies are open-source, cost-effective, and well-supported online.

- *Skillset*

The project team has experience with modern web technologies such as:

- React
- Node.js
- MySQL API integration
- UI/UX design

This ensures efficient development and reduces the learning curve.

➢ *Development Time & Resources*

The project will be developed in phases, allowing the frontend and backend to be built simultaneously.
Given the team's familiarity with web development, the time allocated for each phase is realistic and manageable.

1.7.3 Economic Feasibility

The **Elite Ethiopia platform** is economically feasible to develop, deploy, and maintain. During the final year project phase, development costs are minimal because the system is built using free and open-source technologies and implemented by the project team without the need to hire external developers.

After completion of the final year project, the system is intended to **be** officially released and deployed for real-world use. At this stage, additional operational costs—particularly VPS (Virtual Private Server) hosting, domain services, and routine maintenance—will be incurred. However, these costs remain affordable and scalable, making the system financially sustainable in the long term.

➢ *Cost–Benefit Summary*

- Low development cost due to in-house development.
- Use of open-source tools significantly reduces software expenses.
- Affordable VPS and hosting services suitable for scaling user demand.
- Long-term financial and operational benefits outweigh initial costs.
- Revenue opportunities through service commissions and digital product sales.

- *Cost–Benefit Analysis*

  - ✓ *Tangible Benefits*

- Automates the process of connecting freelancers with clients.
- Reduces fraud and misunderstandings through ratings and dispute management.
- Minimizes time spent on manual communication and coordination.
- Enables digital product creators to sell products directly online.
- Automatically generates financial reports and analytics.
- Provides 24/7 access to digital services nationwide.

  - ✓ *Intangible Benefits*

- Builds trust between service providers and clients.
- Contributes to Ethiopia's digital economy growth.
- Expands employment and income-generation opportunities.
- Promotes transparency and fair competition.
- Encourages youth participation in digital entrepreneurship.
- Enhances the credibility of local digital service providers.

Table 3: Development Phase Cost

Device	Amount	Price	Total
Computers	5	40000	200,000
Sketch books	2	100	200
Wi-Fi pod with unlimited internet	1	6000	6000
Pen and pencil	10	15	150
Papers	100	2	200
Printing costs	20	10	200
Total			**206,750.00**

Table 4: Real-World Deployment Cost

NO.	Item	Quantity	Estimated Cost (ETB)	Total Cost (ETB)
1	Domain & Hosting	1 year	4500	4500
2	VPS Hosting Service	1 year	16,830	16,830
Total				**21,330 ETB**

The project costs are divided into development phase costs and real-world deployment costs to clearly distinguish academic expenses from operational expenses.

The **development phase cost** includes expenses related to printing project documents, computer usage, internet access, and stationery materials. These costs are incurred during the system analysis, design, implementation, and documentation stages of the final year project.

The **real-world deployment cost** represents the expenses required after the system is officially released for public use. This mainly includes domain registration, hosting services, and VPS hosting to ensure system availability, performance, and scalability.

Overall, the cost structure demonstrates that the Elite Ethiopia system is **economically feasible**, with reasonable development expenses and manageable deployment costs that support long-term sustainability.

1.7.4 Behavioral / Political Feasibility

- *Behavioral Feasibility*

Introducing a new digital marketplace may initially face resistance from users unfamiliar with online systems. Some freelancers or clients may be hesitant due to:

- Lack of experience with digital platforms
- Fear of scams or misinformation
- Difficulty adapting to structured processes

To address this, the system will:

- Provide tutorials and user guides
- Use a simple and intuitive interface
- Offer customer support features
- Gradually introduce users to platform features
- Build trust through transparent reviews and dispute resolution

Over time, as users experience its convenience, the platform is expected to be widely accepted.

➢ *Political Feasibility*

The system aligns with Ethiopia's national goals of digital transformation and job creation. It supports:

- Youth employment
- Digital entrepreneurship
- Financial inclusion through local payment systems
- Innovation in ICT

Since the platform does not violate any government policy and works according to digital commerce and financial regulations, it is politically acceptable.

Additionally, universities, startups, and tech communities are likely to support the platform's mission.

1.7.4.3 Schedule Feasibility

Table 5: Schedule Feasibility

Task Name	Timeline
Proposal Preparation	December 2 – December 12
Requirement Analysis	December 13 – January 24
System Design	January 25 – February 28
Implementation	February 29 – March 28

Testing	March 10 – April 15
Project Defense	May

Implementation and testing activities will be carried out in parallel. While system modules are being implemented, testing will begin on completed components to identify and correct errors early. Frontend and backend development will also proceed simultaneously to ensure efficient use of time and timely project completion.

Reasons Why the Schedule Is Feasible

- **Team Expertise:** The team already has background in web development.
- **Parallel Development:** Frontend and backend will be built simultaneously.
- **Realistic Time Allocation:** Each phase has enough time based on complexity.
- **Stakeholder Availability:** Advisor meetings and feedback sessions scheduled early.
- **Modular Development:** Features are built in manageable, smaller modules.

1.8 Scope of the Project

The scope of this project focuses on designing and developing a **public web-based platform** that enables users to offer services, hire freelancers, and buy or sell digital products within Ethiopia.

➢ *Functional Scope*

The system includes:

1. **Freelancing Service Marketplace**

    - Creating, posting, editing, and deleting service offerings
    - Hiring freelancers for specific tasks
    - Submitting service requests and receiving offers
    - Managing orders, negotiations, deadlines, and deliverables

2. **Digital Product Marketplace**

- Uploading and selling digital files (ebooks, templates, videos, software, graphics, audio files)
- Managing product licenses, versions, and downloads
- Secure delivery of purchased digital items

3. **User Management**

- Registration for individuals and organizations
- User profiles, portfolios, and service history
- Role-based access (client, freelancer, admin)

4. **Communication Module**

- Real-time messaging using WebSockets
- File sharing within chat
- Service-related communication channels

5. **Payment Integration**

- Telebirr
- Secure transaction handling
- Commission management and receipts

6. **Ratings & Reviews**

- Rating freelancers after order completion
- Reporting issues or disputes
- Review moderation and verification

7. **Administrative Tools**

- User management

- Product and service moderation
- Financial reporting
- System analytics

➢ *Non-Functional Scope*

- Platform accessibility (desktop + mobile responsive)
- High usability and intuitive navigation
- Moderate scalability for future growth
- Secure authentication and access control
- Data backup and recovery mechanisms

➢ *Out-of-Scope*

- Native Android/iOS mobile apps
- Automated AI-based matching
- Cryptocurrency payments
- International payment support (future version)

1.9 Target Beneficiaries

The system directly benefits:

- Freelancers (designers, writers, developers, marketers, musicians, etc.)
- Digital content creators
- Businesses and organizations seeking professional digital services
- Young graduates searching for online income opportunities
- Clients needing affordable and reliable freelancers
- Administrators responsible for digital marketplace management

Indirect beneficiaries include the broader Ethiopian economy and digital service ecosystem.

1.10 Methodology for the Project

The development of the Elite Ethiopia web-based system followed a systematic and structured methodology grounded in established software engineering principles. The methodology was designed to ensure that the system effectively addresses real-world problems related to online service delivery, digital product exchange, and electronic payments. Emphasis was placed on understanding user needs, analyzing existing practices, and designing a system that is reliable, secure, and scalable.

This methodology integrated data collection, requirement analysis, system modeling, and validation activities. By applying multiple data sources and fact-finding techniques, the project minimized ambiguity and ensured that the resulting system design was both technically sound and user-centered.

1.10.1 Data Source

In order to collect accurate, reliable, and relevant information for the development of the Elite Ethiopia system, the project utilized both **primary and secondary data sources**. The combination of these data sources ensured a comprehensive understanding of the problem domain and helped validate system requirements from multiple perspectives.

- *Primary Data Source*

Primary data were collected directly from individuals and groups who are potential users or stakeholders of the Elite Ethiopia system. These included freelancers, service providers, digital product creators, customers, students, and small business owners. The primary data focused on understanding:

- Existing methods of offering and purchasing services online
- Challenges faced in online payments and service delivery
- Trust and transparency issues between service providers and clients
- Communication and dispute resolution problems
- User expectations regarding system usability and security

Primary data sources were critical because they provided first-hand information about real-life problems and user needs within the local context. This data directly influenced the identification of system features such as secure payment handling, service listing, digital product delivery, messaging, and administrative monitoring.

- *Secondary Data Source*

Secondary data were collected from existing materials that provided background knowledge and technical guidance for the project. These sources included:

- Academic textbooks and lecture materials on software engineering and system analysis
- Research articles related to Elite Ethiopia systems and e-commerce platforms
- Online resources and case studies of similar platforms such as Fiverr, Upwork, and Freelance.com

Secondary data helped the project team understand best practices, standard system architectures, security requirements, and implementation strategies. These sources were also used to compare features and avoid common design and implementation errors.

1.10.2 Fact-Finding Techniques

To gather complete and accurate system requirements, the project employed multiple fact-finding techniques. Using more than one technique allowed cross-verification of information and reduced the risk of missing critical requirements.

1.10.2.1 Interview

Interviews were one of the primary fact-finding techniques used in this project. Both structured and semi-structured interviews were conducted with potential users and stakeholders of the Elite Ethiopia system. The interview sessions aimed to gather in-depth qualitative information that could not be easily obtained through surveys or observation alone.

The interviews focused on the following areas:

- Types of services and digital products users intend to offer or purchase
- Problems encountered with existing online platforms and informal sales channels
- Preferred payment methods and concerns regarding online transactions
- Expectations related to service quality, delivery timelines, and dispute handling
- User opinions on platform trust, transparency, and rating systems

Interviews provided detailed insights into user behavior, expectations, and pain points. The information collected helped shape core system requirements such as secure user registration, escrow-based payment processing, integrated messaging, and review mechanisms.

1.10.2.2 Practical Observation

Practical observation was used to study how service providers and customers currently interact in real-life and online environments. This technique involved observing existing practices without directly interfering with the process.

The team observed how users currently:

- Advertise services and digital products through social media platforms
- Communicate with clients using messaging applications
- Negotiate prices and service requirements manually
- Handle payments without formal verification
- Resolve disputes informally or abandon transactions

Through observation, several inefficiencies were identified, including lack of structured workflows, absence of order tracking, risk of fraud, and poor documentation. These findings were crucial in identifying gaps that the proposed Elite Ethiopia system aimed to address.

Practical observation helped validate information gathered from interviews and ensured that the system design aligned with actual user behavior rather than assumptions.

1.10.2.3 Document Analysis

Document analysis was conducted to review existing written materials relevant to the project. This technique helped ensure that the system complied with academic standards, technical guidelines, and industry practices.

Documents analyzed included:

- University project guidelines and documentation standards
- Sample SRS documents from previous senior projects
- System manuals and process descriptions of similar platforms
- Payment gateway technical documentation and integration guides
- Legal and policy documents related to digital services and online transactions

By analyzing these documents, the project team gained a clearer understanding of required system components, documentation structure, and compliance requirements. Document analysis also helped standardize terminology and align the project with accepted software engineering practices.

1.10.3 Systems Analysis and Design

The systems analysis and design phase was a critical stage in the development of the Elite Ethiopia web-based system. This phase focused on transforming the collected requirements into a structured, logical, and technical representation of the system. The objective was to clearly understand how the system would function, how data would flow, and how different system components would interact with one another before actual development began.

This phase helped minimize development risks, reduce ambiguity, and ensure that the final system met both functional and non-functional requirements.

1.10.3.1 System Analysis

System analysis involved a detailed examination of the problem domain and the proposed solution. The purpose of this activity was to understand user requirements, define system boundaries, and identify the interactions between users, processes, and data.

- *Identification of System Actors*

During the analysis, the major system actors were identified based on user roles and responsibilities. These included:

- **Client** – users who browse, request, and purchase freelance services or digital products
- **Freelancer** – users who provide professional services to clients through the platform
- **Administrators** – users responsible for system management and monitoring
- **Digital Product Creator** – users who upload, manage, and sell digital products on the platform

Each actor was analyzed in terms of permissions, responsibilities, and system interactions.

- *Requirement Analysis*

Functional and non-functional requirements were analyzed in detail. Functional requirements focused on what the system should do, such as:

- User registration and authentication
- Service and digital product management
- Online ordering and payment processing
- Messaging and notifications
- Rating, review, and dispute handling

Non-functional requirements focused on system quality attributes, including:

- Security
- Performance
- Scalability
- Reliability
- Usability

All requirements were documented, validated, and prioritized to guide system design.

- *Process and Data Analysis*

To understand how data moves within the system, process modeling techniques were used. Data Flow Diagrams (DFDs) were developed at different levels to illustrate:

- Input and output data
- Data stores
- Data processing steps
- Interactions between users and the system

These diagrams helped ensure that data handling was efficient, consistent, and secure.

1.10.3.2 System Design

System design translated the results of the analysis into a technical blueprint for implementation. The design phase focused on how the system would be built, rather than what it would do.

- *Architectural Design*

The Elite Ethiopia system was designed using a layered architecture to improve maintainability, scalability, and security. The architecture was divided into the following layers:

- **Presentation Layer:** Handles user interaction through a web interface
- **Application Layer:** Contains business logic and validation rules
- **Data Layer:** Manages data storage and retrieval
- **Integration Layer:** Handles communication with external services such as payment gateways and notification systems

This separation of concerns ensured that changes in one layer would not significantly impact others.

- *Database Design*

A structured database design was created to store system data securely and efficiently. The design included:

- Identification of entities such as users, services, products, orders, payments, and reviews
- Definition of relationships between entities
- Specification of primary keys, foreign keys, and constraints
- Normalization to reduce redundancy and improve data integrity

An Entity Relationship Diagram (ERD) was developed to visually represent the data model.

➢ *User Interface Design*

The user interface was designed with a focus on usability, simplicity, and responsiveness. Wireframes and screen mockups were created for major system pages, including:

- Home page
- User dashboard
- Service and product listing pages
- Order and payment pages
- Messaging interface
- Administrative dashboard

The design ensured consistency in layout, color scheme, and navigation to enhance user experience.

➢ *Security Design*

Security considerations were incorporated at the design stage to protect user data and system resources. The security design included:

- Role-based access control
- Secure authentication mechanisms
- Data validation and error handling
- Protection against unauthorized access and common web attacks

- *Output of the Design Phase*

The system design phase produced several key deliverables, including:

- Use case diagrams and descriptions
- Data Flow Diagrams
- Entity Relationship Diagrams
- System architecture diagrams
- UI mockups and design guidelines

These deliverables served as a reference throughout development and testing.

1.10.4 Development Tools

The development of the Elite Ethiopia web-based system required the use of reliable, efficient, and widely supported software development tools and technologies. The selected tools were chosen based on their availability, cost-effectiveness, scalability, performance, and the development team's familiarity with them. Using modern development tools ensured that the system could be implemented efficiently while maintaining high quality, security, and maintainability.

The tools were categorized into front-end development, back-end development, database management, development environments, testing tools, and deployment-related tools.

Table 6: Development tools

No.	Tool / Technology	Category	Purpose / Description
1	React.js/ NextJS	Front-End Framework	Used to build reusable UI components and single-page application features
2	Node.js	Back-End	Provides server-side runtime environment
3	Express.js	Back-End Framework	Used to build RESTful APIs and manage server routing

4	MySQL	Database	Stores user data, services, orders, payments, and reviews
5	Visual Studio Code	IDE	Used for writing, editing, and debugging code
6	Git	Version Control	Tracks changes in source code and supports collaboration
7	GitHub	Repository Hosting	Stores project code and manages version history
8	Postman	Testing Tool	Used to test and validate API endpoints
9	Web Browsers (Chrome, Firefox)	Testing	Used to test UI functionality and compatibility
10	Payment Gateway APIs (Telebirr)	Integration	Enables secure online payment processing
11	Hosting Platform	Deployment	Hosts the web application and database
12	SSL Certificate	Security	Ensures secure communication using HTTPS

1.10.5 Testing Procedures

Testing was conducted throughout the development of the Elite Ethiopia web-based system to ensure that the system meets its functional requirements, performs reliably, and provides a secure and user-friendly experience. The testing process aimed to identify and correct errors early, validate system behavior, and ensure that all components work together as expected.

- *Unit Testing*

Unit testing was performed on individual modules of the system to verify that each component functions correctly in isolation. Modules such as user registration, login, service listing, digital product upload, and payment processing were tested separately. This helped detect logical errors at an early stage of development.

- *Integration Testing*

Integration testing was conducted to ensure that different system modules interact correctly with one another. For example, integration between the payment system and order management module, as well as between the messaging system and user accounts, was tested. This ensured smooth data flow and correct communication between components.

- *System Testing*

System testing was carried out after integrating all system components. The complete system was tested to verify that it meets all specified requirements. This included testing end-to-end workflows such as service ordering, payment completion, service delivery, and review submission.

- *User Acceptance Testing (UAT)*

User acceptance testing involved selected users testing the system in real-world scenarios. Feedback was collected regarding usability, clarity, and performance. This helped confirm that the system meets user expectations and is ready for deployment.

- *Security Testing*

Security testing was conducted to protect user data and system resources. Authentication mechanisms, access control, and input validation were tested to prevent unauthorized access and common security threats.

1.10.6 Implementation Method

Implementation refers to the process of deploying the new system and transitioning users from existing practices to the new Elite Ethiopia platform. Three common implementation approaches were considered: **Parallel, Partial, and Direct implementation**.

- *Parallel Implementation*

In parallel implementation, the new system runs alongside the existing system for a certain period. Users continue using the old method while also testing the new system. This approach reduces risk, as the old system remains available if problems arise in the new system.

- *Partial Implementation*

Partial implementation involves introducing the new system gradually by deploying specific modules or features first. For example, user registration and service listing may be deployed before payment and dispute resolution features. This approach allows gradual adaptation and controlled risk.

- *Direct Implementation*

In direct implementation, the old system is completely replaced by the new system at a specific point in time. This approach is fast but carries higher risk, as any failure in the new system can disrupt operations.

- *Implementation Approach Used*

For this project, a **parallel implementation approach** was selected. The Elite Ethiopia system was introduced while users continued using existing informal platforms such as social media and manual payment methods. This allowed users to familiarize themselves with the new system while reducing the risk of system failure. Feedback gathered during this period was used to improve system stability and usability before full adoption.

1.10.7 Limitation of the Project

Despite careful planning and execution, the project may face several limitations:

- Limited financial resources restricted the use of advanced cloud infrastructure and paid third-party services.
- Dependence on third-party payment gateways introduced constraints related to availability and policy changes.
- Time limitations affected the implementation of advanced features such as AI-based recommendations and advanced analytics.
- Internet dependency may limit system accessibility in areas with poor or unstable connectivity.
- The system was developed within an academic timeframe, which limited large-scale real-world testing.

Chapter Two: Description of the Existing System

2.1 Introduction of Existing System

Before the development of the proposed *Elite Ethiopia* platform, freelancing and digital product transactions in Ethiopia are handled through **fragmented, informal, and inefficient systems**. There is **no single centralized online platform** specifically designed to support Ethiopian freelancers, digital creators, and clients using **local payment systems**.

Currently, individuals offering digital services such as graphic design, software development, writing, video editing, music production, and digital marketing rely heavily on **social media platforms** (Facebook, Telegram, WhatsApp, Instagram), personal referrals, or informal online communities to find clients. Similarly, digital creators who wish to sell products such as eBooks, digital art, templates, music, or software distribute their products manually through chat-based communication and file-sharing tools.

International freelancing platforms like **Fiverr, Upwork, and Freelancer.com** exist, but they are **largely inaccessible** to Ethiopian users due to:

- Lack of support for Ethiopian payment systems
- Restrictions on account creation
- Difficulty withdrawing earnings
- Currency and international banking limitations

As a result, most transactions in the existing system are conducted **manually**, without structured workflows, automated tracking, secure payment escrow, or formal dispute resolution mechanisms. Communication, negotiation, delivery, and payment confirmation all depend on trust between individuals, which often leads to misunderstandings, delays, and fraud.

The absence of a formal digital marketplace tailored to Ethiopia limits the growth of digital entrepreneurship, reduces income opportunities for skilled individuals, and creates inefficiencies for businesses seeking reliable service providers. This existing system is therefore inadequate to

support the growing digital economy and demands a more organized, secure, and localized solution.

2.2 Players in the existing system

The existing system involves several key players who interact in an unstructured and informal environment. These players include the following:

- *Freelancers / Service Providers*

Freelancers are individuals who offer digital services such as programming, graphic design, content writing, digital marketing, video editing, and music production. In the existing system:

- They advertise services through social media posts or personal networks
- Negotiations are conducted via chat or phone calls
- Payments are requested manually using local transfer methods
- There is no formal contract, rating system, or service tracking mechanism

- *Digital Product Creators*

Digital product creators develop and sell intangible products such as:

- eBooks
- Digital artwork
- Templates
- Software
- Audio and video files

They distribute their products through:

- Telegram channels
- Facebook pages
- Direct messaging
- File-sharing platforms

There is no secure delivery system, license control, or automated download management, making piracy and payment disputes common.

- *Clients / Service Buyers*

Clients include individuals, businesses, and organizations seeking digital services or products. In the existing system:

- Clients search for providers manually through social media or referrals
- They lack verified information about service quality
- There is no rating or review system to guide decision-making
- Payment confirmation and service completion rely solely on trust

- *Financial Service Providers*

Local payment systems such as:

- Telebirr

Are used independently of any centralized system. Payments are made manually, without automated verification, transaction tracking, or escrow protection.

- *Platform Administrators (Informal)*

In some cases, administrators of Telegram groups or Facebook pages act as informal moderators by:

- Approving posts
- Removing spam
- Resolving conflicts informally

However, they lack authority, accountability, system tools, and structured rules.

2.3 Major functions in the existing system

The system provides an integrated web-based environment that supports freelancing services and digital product transactions. All system activities, including communication, service management,

payments, and reporting, are performed internally within the platform. The major functions of the system are described in detail as follows.

- *User Registration and Authentication*

The system allows users to register as individuals or organizations by providing required personal or business information. During registration, users create secure login credentials. The system authenticates users during login to ensure that only authorized users can access platform features. Role-based access control is applied to distinguish between clients, freelancers, digital product creators, and administrators.

- *Real-Time Communication and Collaboration*

The system provides an integrated real-time texting feature that enables direct communication between users. This messaging module allows clients, freelancers, and digital product creators to exchange messages instantly within the platform.

The communication feature supports:

- Clarification of service requirements
- Negotiation of pricing and delivery timelines
- Collaboration during service execution
- Communication related to digital product inquiries and support

All messages are stored securely in the system database and are linked to specific users, services, or orders. This ensures communication traceability, accountability, and availability for future reference or dispute resolution.

- *Service Posting and Management*

The system enables freelancers to create and manage service listings. Each service listing includes structured information such as service title, detailed description, category, pricing, delivery time, and revision policies.

Freelancers can update or deactivate their services as needed. The system ensures that all service data is stored in a standardized format, allowing clients to easily browse, search, and compare services.

- *Service Request and Order Processing*

Clients can browse available services and submit service requests directly through the platform. Once a service is selected, the system creates an order that records all relevant details, including service scope, price, delivery time, and assigned service provider.

The system tracks the status of each order from initiation to completion, providing visibility to both clients and freelancers throughout the service lifecycle.

- *Payment Processing and Transaction Management*

The system supports secure payment processing using locally supported payment methods. Payments are initiated and confirmed within the platform. The system records transaction details, including payment status, amount, date, and payer information.

Access to services or digital products is granted only after successful payment confirmation. This ensures transactional security and financial transparency.

- *Digital Product Upload and Distribution*

The system allows digital product creators to upload digital products such as eBooks, software, templates, audio, or video files. Each product includes metadata such as title, description, price, category, and version information.

Upon purchase, the system securely delivers the digital product to the buyer through controlled download access. The system manages download availability and maintains records of product purchases.

- *Service Delivery and Completion Management*

Freelancers submit completed service deliverables through the platform. Clients review the submitted work and confirm acceptance. Once accepted, the system marks the service as completed.

This function ensures a clear and documented service completion process, reducing ambiguity and disputes.

- *Rating and Feedback Management*

After service or product completion, the system allows clients to provide ratings and written feedback. These ratings are associated with the corresponding freelancer or product and are displayed to other users.

The feedback system supports transparency, quality assurance, and informed decision-making.

- *Administrative Management and Monitoring*

The system provides administrative functions for managing users, services, products, payments, and system activity. Administrators can monitor platform usage, resolve disputes, moderate content, and generate reports.

2.4 Business rules

- *User Roles and Permissions*

The system enforces role-based access control to regulate user activities. Every registered user is assigned a predefined role such as client, freelancer, digital product creator, or administrator. Each role determines the operations that the user is authorized to perform within the system. Users are restricted from accessing features that fall outside their assigned roles, ensuring controlled and secure system operation.

- *Communication Rules*

All communication between users must be conducted through the system's integrated real-time messaging feature. Messages exchanged within the platform are considered official communication and are securely stored in the system database. These communication records are associated with specific users and transactions, allowing traceability, accountability, and effective dispute resolution.

- *Service and Product Posting Rules*

Only authorized users are permitted to create service listings or upload digital products. All listings must include complete and accurate information such as descriptions, pricing, and delivery conditions. Once a service or product is purchased, its core terms cannot be modified unless changes are mutually agreed upon through the system.

- *Payment and Transaction Rules*

All financial transactions must be processed through the platform using supported payment methods. Services and digital products are not delivered until payment is successfully completed and verified. Transaction details, including payment status and timestamps, are stored permanently to ensure transparency and auditability.

- *Service Delivery and Completion Rules*

Service providers are required to deliver services in accordance with the agreed scope, pricing, and delivery time. Clients must review submitted deliverables and confirm acceptance through the system. A service is considered complete only after client approval or system-defined completion conditions are met.

- *Rating, Review, and Dispute Rules*

Only users who have completed a transaction are allowed to submit ratings and reviews. In case of disputes, complaints must be submitted through the system. Administrators review disputes

using system records, including communication logs and transaction histories, and resolve issues based on platform policies.

2.5 Reports Generated by the System

- *Transaction Reports*

The system generates transaction reports that provide detailed information on completed payments, pending transactions, refunds, and commission summaries. These reports support financial monitoring and decision-making.

- *Service Performance Reports*

Service performance reports summarize service-related activities such as completed orders, delivery timelines, and customer ratings. These reports help evaluate service quality and provider performance.

- *Digital Product Sales Reports*

The system generates digital product sales reports that include information on product listings, total sales, revenue generated, and download statistics. These reports assist creators in analyzing product performance.

- *User Activity Reports*

User activity reports record information related to user registration, login behavior, and platform usage. These reports support system monitoring and security analysis.

- *Administrative and Audit Reports*

Administrative and audit reports document moderation activities, dispute resolutions, system events, and security logs. These reports ensure accountability and operational transparency.

2.6 Forms and Other Documents Used in the System

- *User Registration Forms*

The system provides structured registration forms for individuals and organizations. These forms collect essential user information and include validation mechanisms to ensure data accuracy.

- *Service and Digital Product Listing Forms*

Standardized forms are used to create service and digital product listings. These forms ensure consistent data entry and presentation across the platform.

- *Order and Transaction Documents*

For every service order or product purchase, the system generates digital order and transaction documents. These documents record order identifiers, user details, and payment status.

- *Payment Receipts and Invoices*

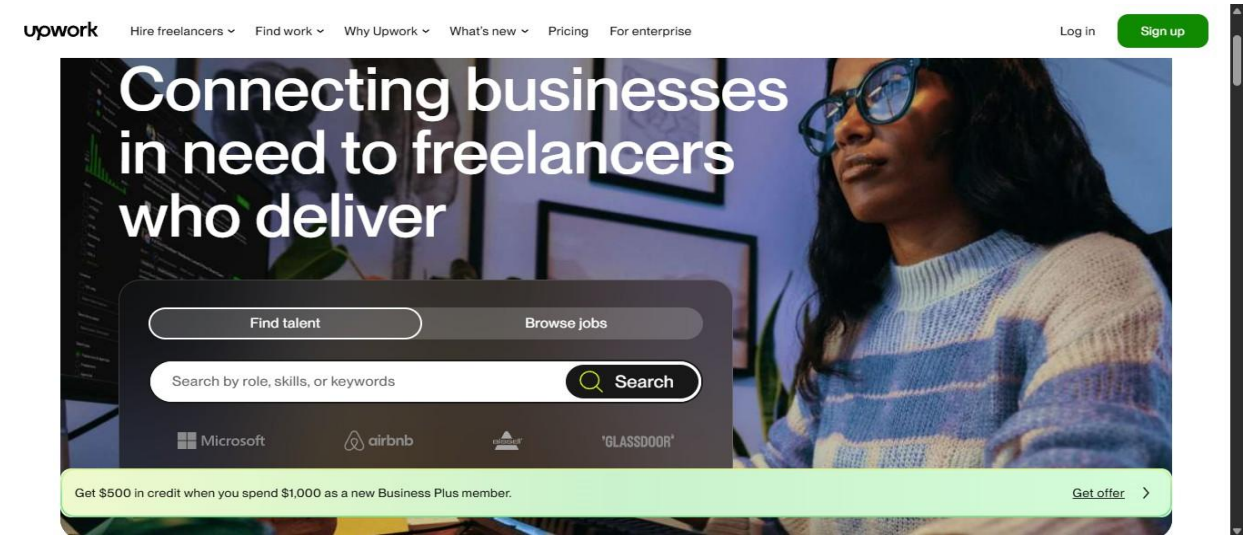
Payment receipts and invoices are automatically generated after successful transactions. These documents serve as official proof of payment and are stored within the system.

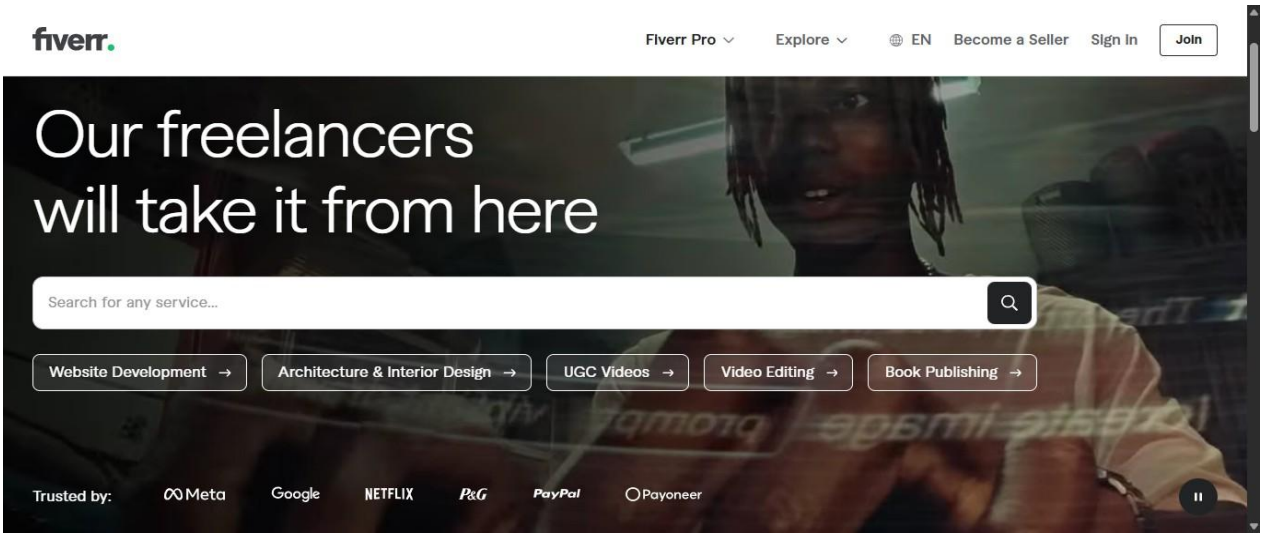
- *Communication Records*

All real-time communication messages exchanged within the platform are stored as official system records. These records are linked to user accounts and transactions for reference and dispute handling.

- *System Logs and Reports*

The system maintains operational logs and system-generated reports, including activity logs and security logs, to support monitoring, auditing, and maintenance activities.





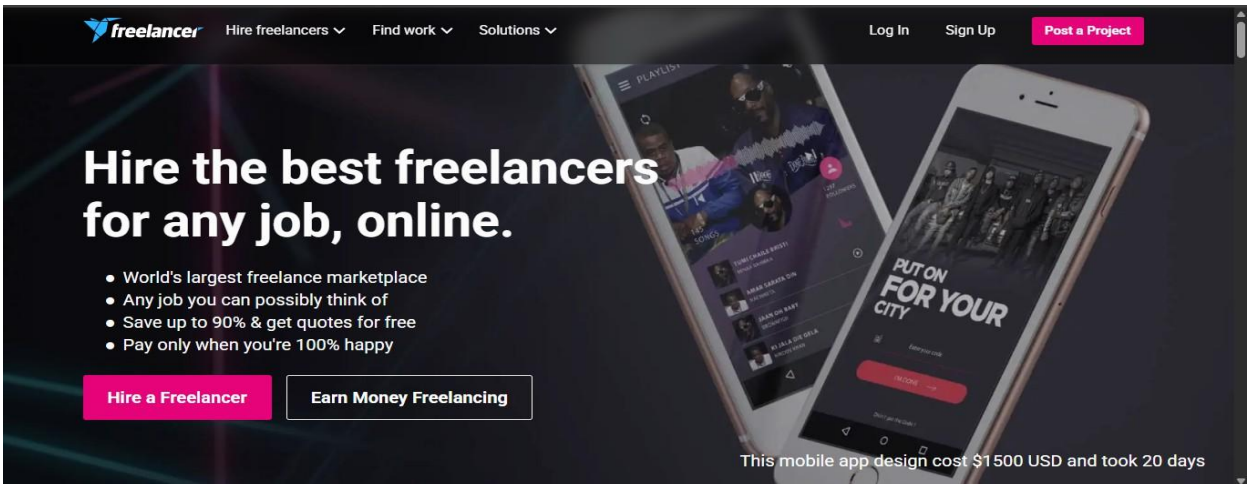


Figure 1: Forms, Interfaces and Other Information of the Existing System

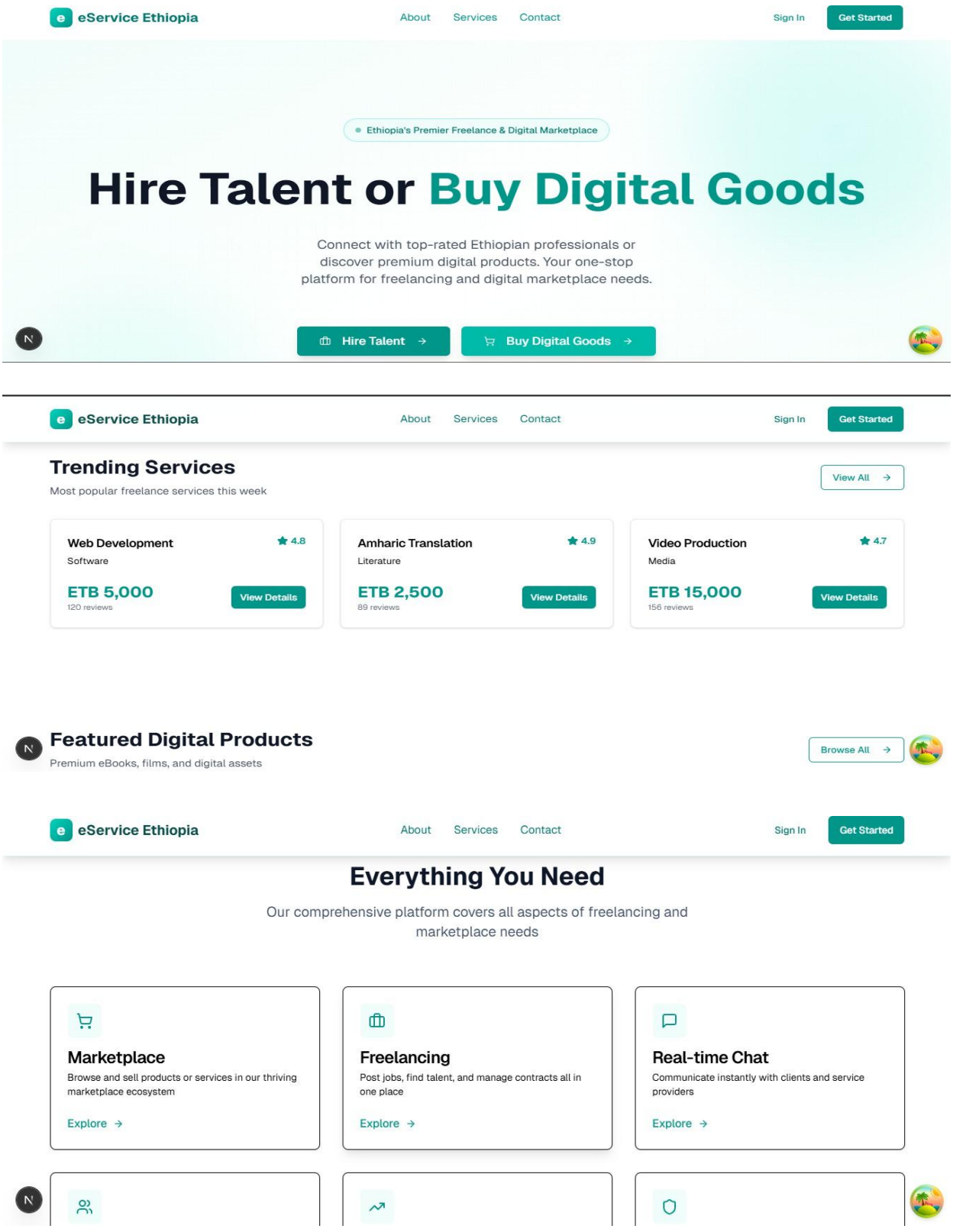


Figure 2: Elite Ethiopia informative site:

2.7 Bottlenecks of the System

Despite providing integrated digital services, the system has several limitations that affect its performance, usability, security, and efficiency. These bottlenecks highlight areas that require improvement and optimization.

2.7.1 Performance (Response Time)

As the number of users and transactions increases, the system may experience performance challenges such as slower response times during peak usage. Real-time communication, file uploads, and payment processing require continuous server resources. Without optimized backend processing, database indexing, and load handling mechanisms, delays may occur when multiple users access the system simultaneously.

2.7.2 Input and Output Limitations

The system relies on user-provided input for service descriptions, product details, and communication messages. Inaccurate or incomplete input can affect service quality and user satisfaction. Additionally, without strict validation rules, redundant or inconsistent data may be entered into the system. Output-related limitations include unclear presentation of service status, transaction feedback, or report summaries if not properly formatted or updated in real time.

2.7.3 Security and Control Issues

The system handles sensitive data such as user credentials, communication records, and financial transactions. Without strong security controls, the system may be vulnerable to unauthorized access, data breaches, or misuse of user accounts. Potential security bottlenecks include weak authentication mechanisms, insufficient access control, and lack of monitoring for suspicious activities.

2.7.4 Efficiency Constraints

Manual administrative processes such as service moderation, dispute resolution, and report review can reduce operational efficiency. As platform usage grows, administrative workload may

increase, leading to delays in approvals, responses, or system maintenance if automation and prioritization mechanisms are not adequately implemented.

2.8 Practices to Be Preserved

While improvements are necessary, several existing system practices provide value and should be preserved in future enhancements.

- *Integrated Real-Time Communication*

The system's built-in real-time messaging feature enables direct and structured communication between users. This practice supports collaboration, reduces misunderstandings, and ensures communication traceability. Maintaining and enhancing this feature is essential.

- *Centralized Platform Operation*

All system activities—service management, communication, payment processing, and reporting—are performed within a single platform. This centralized approach simplifies user experience and improves data consistency. This practice should be preserved and strengthened.

- *Local Payment System Support*

The system's support for locally available payment methods enables accessibility and financial inclusion. Preserving local payment integration ensures that users can transact conveniently and securely.

- *Structured User Roles*

The role-based access model ensures that users interact with the system according to defined permissions. This practice improves security and operational clarity and should remain a core system feature.

2.9 Proposed Solution for the New System

To address the identified bottlenecks and enhance system effectiveness, a refined and improved version of the system is proposed.

- *System Optimization and Scalability*

The proposed solution includes optimizing backend services, improving database performance, and implementing scalable architecture. These improvements will enhance response time and allow the system to handle increased user activity efficiently.

- *Enhanced Input Validation and Data Management*

The new system will enforce strict input validation rules and standardized data formats. This will reduce errors, eliminate redundant data, and improve output accuracy. Clear feedback messages will be provided to users during data entry and transactions.

- *Strengthened Security Mechanisms*

The proposed system will implement stronger security controls, including secure authentication, encrypted data transmission, and role-based authorization. Continuous monitoring and logging mechanisms will be added to detect and prevent unauthorized activities.

- *Improved Administrative Automation*

Administrative processes such as service approval, dispute handling, and report generation will be enhanced through automation. This will reduce manual workload, improve response time, and increase operational efficiency.

- *User Experience Enhancement*

The new system will focus on improving usability through intuitive interfaces, responsive design, and clear navigation. Enhancing user experience will increase adoption and satisfaction among all user groups.

2.10 Requirements of the Proposed System

The proposed system is required to meet a comprehensive set of functional and non-functional requirements to ensure effective operation, reliability, security, and usability. These requirements define the expected behavior of the system as well as the quality attributes it must maintain during real-world operation. Meeting these requirements ensures that the system fulfills user needs while supporting future scalability and maintainability.

2.10.1 Functional Requirements

Functional requirements describe the specific services, tasks, and operations that the system must perform to support users and administrators.

2.10.1.1 User Registration and Authentication

The system shall provide a secure user registration mechanism that allows individuals and organizations to create accounts by supplying valid and complete information. The system shall authenticate users using secure credentials during login and prevent unauthorized access. Account verification mechanisms shall be applied to ensure authenticity and system integrity.

2.10.1.2 User Role Management

The system shall support multiple predefined user roles, including client, freelancer, digital product creator, and administrator. Each role shall have clearly defined permissions that control access to system features. Role-based access control ensures that users can only perform actions relevant to their responsibilities, thereby enhancing security and operational clarity.

2.10.1.3 User Profile Management

The system shall allow users to create and manage detailed profiles containing personal information, skills, service offerings, portfolios, and transaction history. Profile information shall be editable while maintaining data consistency. User profiles provide transparency and help clients make informed decisions when selecting service providers.

2.10.1.4 Real-Time Communication

The system shall provide an integrated real-time communication module that enables users to exchange messages directly within the platform. This feature supports collaboration, negotiation, clarification of requirements, and project coordination. All communication records shall be securely stored and linked to related services or transactions for traceability and dispute resolution.

2.10.1.5 Service Posting and Management

The system shall allow freelancers to create, update, and manage service listings through structured forms. Each service listing shall contain essential information such as service description, pricing, delivery timeframe, and terms. This ensures standardization and consistency across all service offerings on the platform.

2.10.1.6 Service Search and Browsing

The system shall provide users with tools to browse and search services efficiently using categories, keywords, and filters. Search results shall be presented in an organized manner to enhance discoverability and user experience. This functionality helps clients quickly identify services that match their needs.

2.10.1.7 Service Ordering and Tracking

The system shall allow clients to place service orders through a defined workflow. Each order shall be tracked throughout its lifecycle, including request submission, acceptance, delivery, and completion. Order tracking improves transparency and accountability between clients and service providers.

2.10.1.8 Digital Product Upload and Management

The system shall allow authorized users to upload digital products along with required metadata such as title, description, pricing, and version details. Uploaded products shall be securely stored and managed within the system. This functionality supports a structured digital product marketplace.

2.10.1.9 Digital Product Purchase and Delivery

The system shall allow users to purchase digital products through the platform. Upon successful payment, the system shall securely deliver the purchased digital content to the buyer. Access controls shall ensure that only authorized buyers can download purchased products.

2.10.1.10 Payment Processing

The system shall integrate secure local payment methods to process transactions. Payments must be verified before services are activated or digital products are delivered. Transaction details shall be recorded for financial transparency, reporting, and audit purposes.

2.10.1.11 Rating and Review Management

The system shall allow users to submit ratings and reviews after completing a transaction. Reviews shall reflect actual experiences and be linked to completed services or purchases. This feature promotes trust, accountability, and quality assurance within the platform.

2.10.1.12 Dispute Management

The system shall provide a structured dispute submission and handling mechanism. Users may raise disputes related to service quality, delivery delays, or payment issues. Administrators shall review disputes using system records and communication logs to ensure fair resolution.

2.10.1.13 Administrative Management

The system shall provide administrators with a comprehensive dashboard to manage users, services, products, transactions, and system reports. Administrative tools shall support moderation, monitoring, and enforcement of platform policies.

2.10.2 Non-Functional Requirements

Non-functional requirements describe the performance, security, and quality attributes that define how well the system operates.

2.10.2.1 Performance Requirements

The system shall provide timely responses to user requests under normal and peak usage conditions. Efficient processing of transactions, communication, and data retrieval is required to maintain a smooth user experience.

2.10.2.2 Security Requirements

The system shall implement strong security mechanisms to protect user data and system resources. Secure authentication, authorization, and data protection practices shall be enforced to prevent unauthorized access and data breaches.

2.10.2.3 Usability Requirements

The system shall provide a clear, intuitive, and user-friendly interface. Navigation and system workflows shall be designed to minimize user effort and reduce the learning curve for new users.

2.10.2.4 Reliability Requirements

The system shall operate reliably with minimal downtime. Data consistency and system stability shall be maintained even during unexpected failures or high usage periods.

2.10.2.5 Scalability Requirements

The system shall be designed to accommodate future growth in users, transactions, and data volume. The architecture shall support feature expansion without major redesign.

2.10.2.6 Maintainability Requirements

The system shall follow modular design principles that simplify maintenance, debugging, and future enhancements. Clear documentation and coding standards shall support long-term maintainability.

Chapter Three: System Analysis

3.1 Introduction

This chapter presents a detailed analysis of the proposed Elite Ethiopia system. System analysis focuses on understanding how the system functions, how users interact with it, and how different components collaborate to achieve the intended objectives. The purpose of this chapter is to transform the requirements identified in the previous chapters into structured models and specifications that clearly describe system behavior.

The chapter emphasizes requirement specification, functional interactions, and conceptual modeling of the system. It introduces various analysis tools such as use case diagrams, sequence diagrams, activity diagrams, class diagrams, and user interface prototypes. These models provide a clear and comprehensive understanding of system operations and serve as a foundation for system design and implementation.

3.2 System Requirement Specifications (SRS)

System Requirement Specification (SRS) defines the functional and non-functional characteristics of the system in a precise and structured manner. It serves as a formal agreement between stakeholders and developers by describing what the system does, how it behaves, and the constraints under which it operates.

The SRS for the Elite Ethiopia platform documents user interactions, system processes, data flow, and operational rules. It ensures clarity, consistency, and completeness of requirements while reducing ambiguity during development and testing. The specification also supports future system maintenance and enhancement by providing a clear reference for system functionality.

The SRS is represented using multiple modeling techniques, including use case diagrams, use case descriptions, sequence diagrams, activity diagrams, class diagrams, and user interface prototypes. Each modeling technique highlights a different perspective of the system to ensure a comprehensive analysis.

3.2.1 Use case diagrams

Use case diagrams are used to represent the functional interactions between users (actors) and the system. They provide a high-level view of the system's functionality by illustrating the services the system offers and the external entities that interact with it. In the Elite Ethiopia system, use case diagrams are used to identify user roles, define system boundaries, and show how different users interact with system functionalities.

The use case diagram of the Elite Ethiopia platform represents the system as a centralized digital marketplace that supports freelancing services and digital product transactions. The diagram clearly separates the system from external entities and shows all major interactions within a single integrated platform.

Actors Identified in the System

The system includes several primary actors, each representing a category of users or external components that interact with the system.

Client

The client represents users who seek services or purchase digital products. Clients interact with the system to browse services, communicate with service providers, place orders, make payments, and provide feedback.

Freelancer

The freelancer represents users who offer professional services through the platform. Freelancers interact with the system to create service listings, communicate with clients, accept orders, deliver services, and receive ratings.

Digital Product Creator

The digital product creator represents users who upload and sell digital products. This actor interacts with the system to manage product listings, monitor sales, and deliver digital content.

Administrator

The administrator represents system managers responsible for overseeing platform operations.

Administrators interact with the system to manage users, approve services and products, monitor transactions, resolve disputes, and generate reports.

Payment Gateway

The payment gateway represents external payment services integrated with the system. This actor interacts with the system during payment processing and transaction verification.

The use case diagram includes major functionalities such as:

- User registration
- Secure login
- Service posting
- Digital product upload
- Order placement
- Online payment
- Messaging and notifications
- Review and rating
- System administration

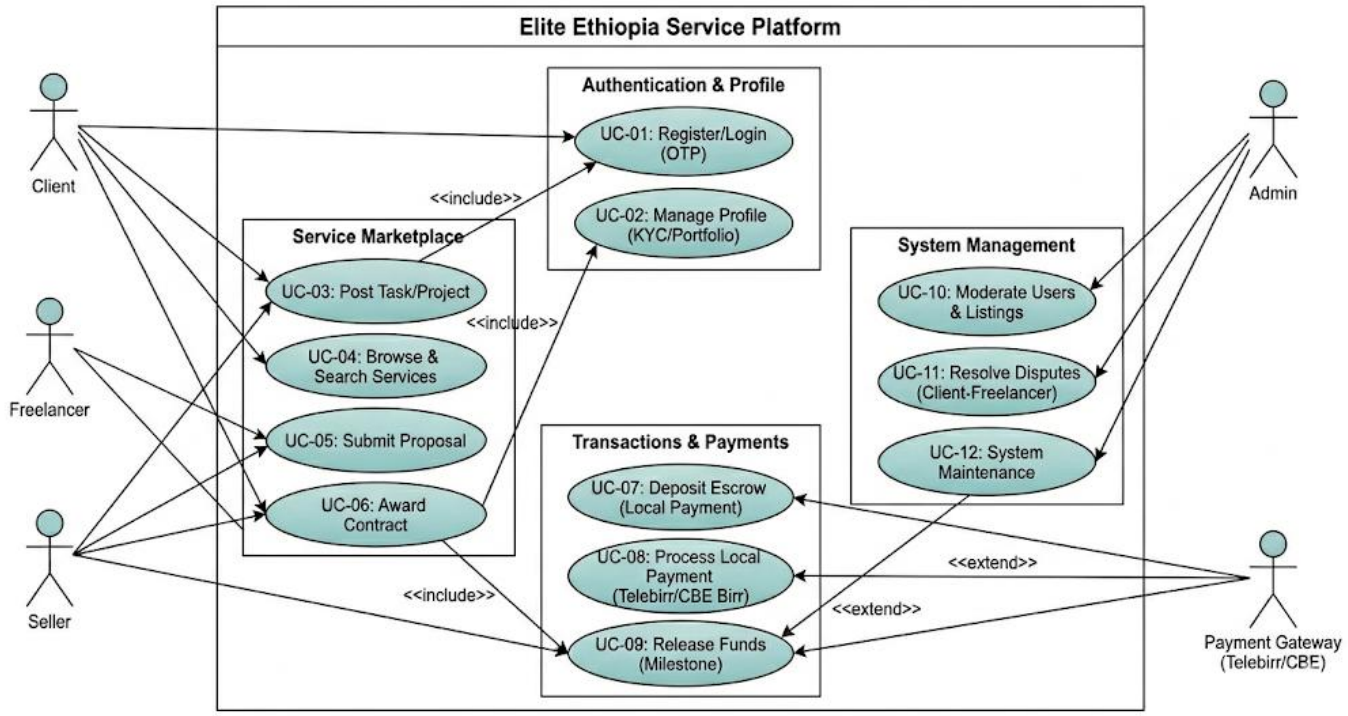


Figure 3: Use case diagram for Elite Ethiopia

3.2.2 Use case documentation

Table 7: Security Login use case description

Field	Description
Use Case Id	UC-01
Use Case Name	Security Login
Actor(s)	Client, Freelancer, Digital Product Creator, Administrator
Description	Ensures secure authentication of all system users to access the Elite Ethiopia platform based on their assigned roles.
Preconditions	1. Users must have valid login credentials (email/username and password). 2. The system must be online and accessible.
Post Conditions	1. Successful login grants access to the user's role-based dashboard. 2. Failed login attempts are recorded for security monitoring.
Normal Flow	1. User navigates to the login page. 2. User enters login credentials. 3. System validates the credentials. 4. On success, the user is redirected to their dashboard.
Alternative Flows	4.1 If credentials are invalid, the system displays an error message and allows the user to retry.
Exceptions	1. System downtime prevents login. 2. Account is locked after multiple failed attempts.
Frequency of Use	Every time a user accesses the system.

Table 8: Registration Use case description

Field	Description

Use Case Id	UC-02
Use Case Name	Registration
Actor(s)	Client, Freelancer, Digital Product Creator
Description	Allows new users to create an account on the Elite Ethiopia platform in order to access system services.
Preconditions	1. The system must be accessible. 2. The user must not already have an account.
Post Conditions	1. User account is successfully created. 2. User information is stored securely in the database.
Normal Flow	1. User navigates to the registration page. 2. User selects account type (Client or Service Provider). 3. User enters required personal and account details. 4. System validates the information. 5. Account is created and confirmation is displayed.
Alternative Flows	4.1 If validation fails (e.g., duplicate email or weak password), the system prompts the user to correct the information.
Exceptions	1. System error prevents registration. 2. Network failure during submission.
Frequency of Use	When a new user joins the platform.

Table 9: Manage Profile Use case description

Item	Description
Use Case ID	UC-03
Use Case Name	Manage Profile
Primary Actors	Client, Freelancer, Digital Product Creator
Description	This use case describes how users update their personal and professional profile information.

Precondition	The user is logged into the system.
Normal Flow	1. The user opens the profile section. 2. The system displays current profile information. 3. The user edits profile details (bio, skills, portfolio, contact info). 4. The system validates and saves the updates. 5. The system confirms profile update.
Alternative Flow	4.1 If invalid data is entered, the system prompts the user to correct it.
Postcondition	The user profile information is updated successfully.

Table 10: Browse & Search ServicesUse case description

Item	Description
Use Case ID	UC-04
Use Case Name	Browse & Search Services
Primary Actor	Client
Description	This use case enables clients to browse and search for available freelance services.
Precondition	The client is logged into the system.
Normal Flow	1. The client accesses the service listing page. 2. The system displays available services. 3. The client searches or filters services by category or keyword. 4. The system displays matching results.
Alternative Flow	4.1 If no matching services are found, the system displays a notification.

Postcondition	The client views available or filtered service listings.

Table 11: Place Service Order Use case description

Item	Description
Use Case ID	UC-05
Use Case Name	Place Service Order
Primary Actor	Client
Supporting Actor	Payment Gateway
Description	This use case describes how a client places an order for a selected service.
Precondition	The client is logged in and has selected a service.
Normal Flow	1. The client selects a service. 2. The system displays service details and pricing. 3. The client confirms the order. 4. The system initiates payment processing. 5. The payment gateway processes the transaction. 6. The system confirms order placement.
Alternative Flow	5.1 If payment fails, the system displays an error and cancels the order.
Postcondition	The service order is successfully placed and recorded.

Table 12: Real-Time Communication Use case description

Item	Description
Use Case ID	UC-06
Use Case Name	Real-Time Communication
Primary Actors	Client, Freelancer

Description	This use case allows users to communicate directly within the platform for collaboration and clarification.
Precondition	Both users are logged in and have an active order or interaction.
Normal Flow	1. The user opens the chat interface. 2. The system establishes a real-time connection. 3. Users exchange text messages and files. 4. The system delivers messages instantly.
Postcondition	Messages are successfully exchanged and stored.

Table 13: Upload & Manage Digital Products Use case description

Item	Description
Use Case ID	UC-07
Use Case Name	Upload & Manage Digital Products
Primary Actor	Digital Product Creator
Description	This use case describes how creators upload and manage digital products for sale.
Precondition	The creator is logged into the system.
Normal Flow	1. The creator selects upload product option. 2. The system displays product upload form. 3. The creator uploads files and enters product details. 4. The system validates and stores the product. 5. The system confirms successful upload.
Postcondition	The digital product becomes available on the platform.

Table 14: Purchase Digital Products Use case description

Item	Description
Use Case ID	UC-08
Use Case Name	Purchase Digital Products
Primary Actor	Client
Supporting Actor	Payment Gateway
Description	This use case enables users to purchase digital products available on the platform.
Precondition	The user is logged in and has selected a digital product.
Normal Flow	1. The user selects a digital product. 2. The system displays product details and price. 3. The user confirms purchase. 4. The system processes payment through the gateway. 5. The system grants product access after successful payment.
Alternative Flow	4.1 Payment failure results in purchase cancellation.
Postcondition	The user gains access to the purchased digital product.

Table 15: Dispute Management

Item	Description
Use Case ID	UC-09
Use Case Name	Dispute Management
Primary Actors	Client, Freelancer, Administrator
Description	This use case handles conflicts related to services or transactions.
Precondition	A completed or active transaction exists.
Normal Flow	1. A user submits a dispute. 2. The system records the dispute. 3. The administrator reviews the case.

	4. The administrator takes appropriate action. 5. The system updates dispute status.
Postcondition	The dispute is resolved and recorded.

Table 16: Administrative Management Use case description

Item	Description
Use Case ID	UC-10
Use Case Name	Administrative Management
Primary Actor	Administrator
Description	This use case allows administrators to manage users, services, products, and system operations.
Precondition	The administrator is logged into the system.
Normal Flow	1. The administrator accesses the admin dashboard. 2. The system displays management options. 3. The administrator performs administrative actions. 4. The system saves changes and updates records.
Postcondition	System data and settings are updated accordingly.

3.2.3 Sequence diagram

Sequence diagrams illustrate the interaction between system components and actors over time. They show the order in which messages are exchanged to accomplish a specific system function. The following sequence diagrams represent the major workflows of the Elite Ethiopia system.

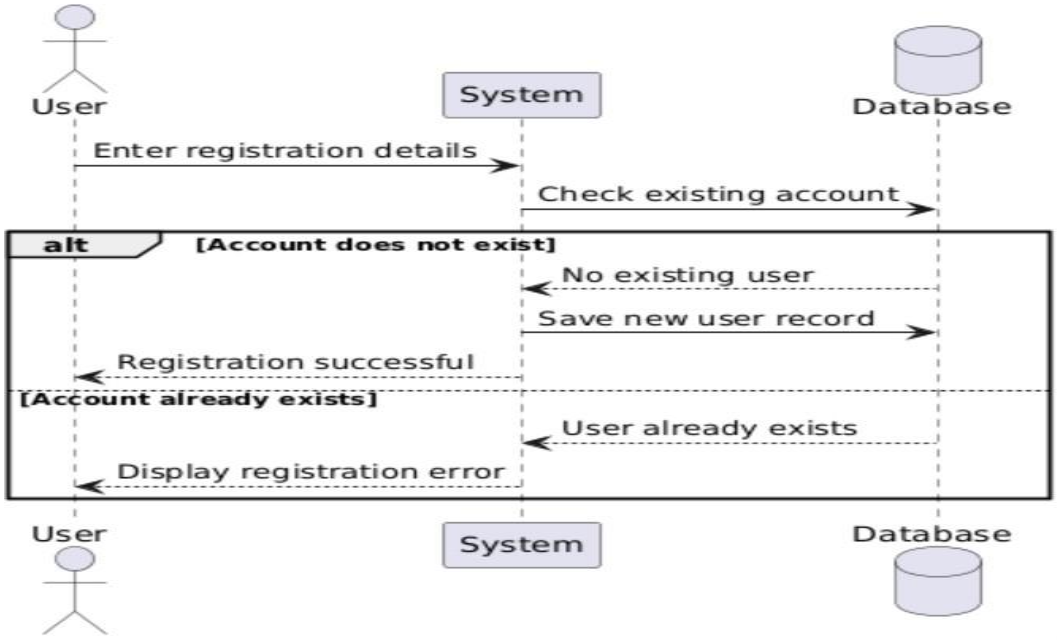


Figure 4: Registration Sequence diagram

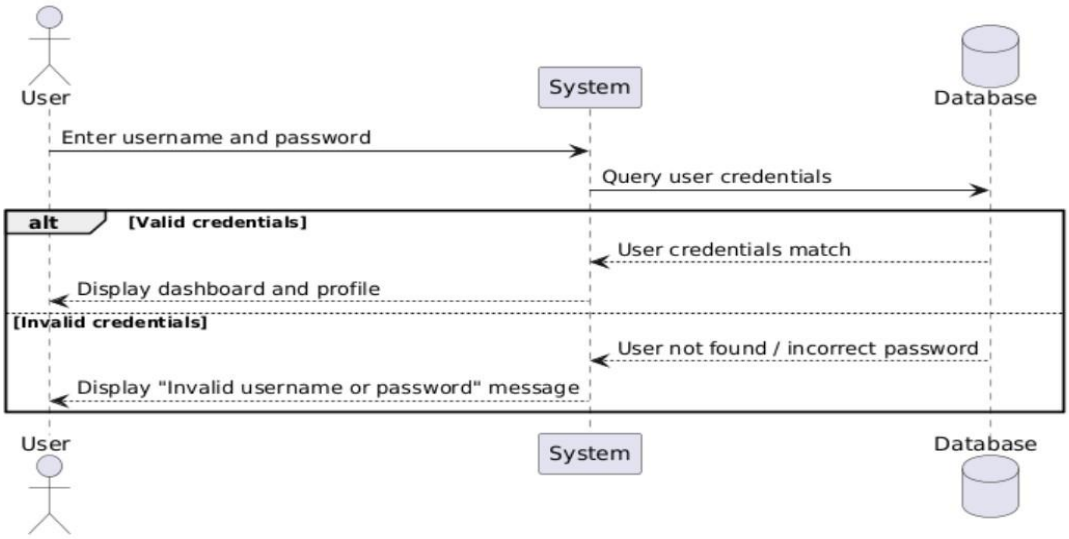


Figure 5: Sequence diagram for Login

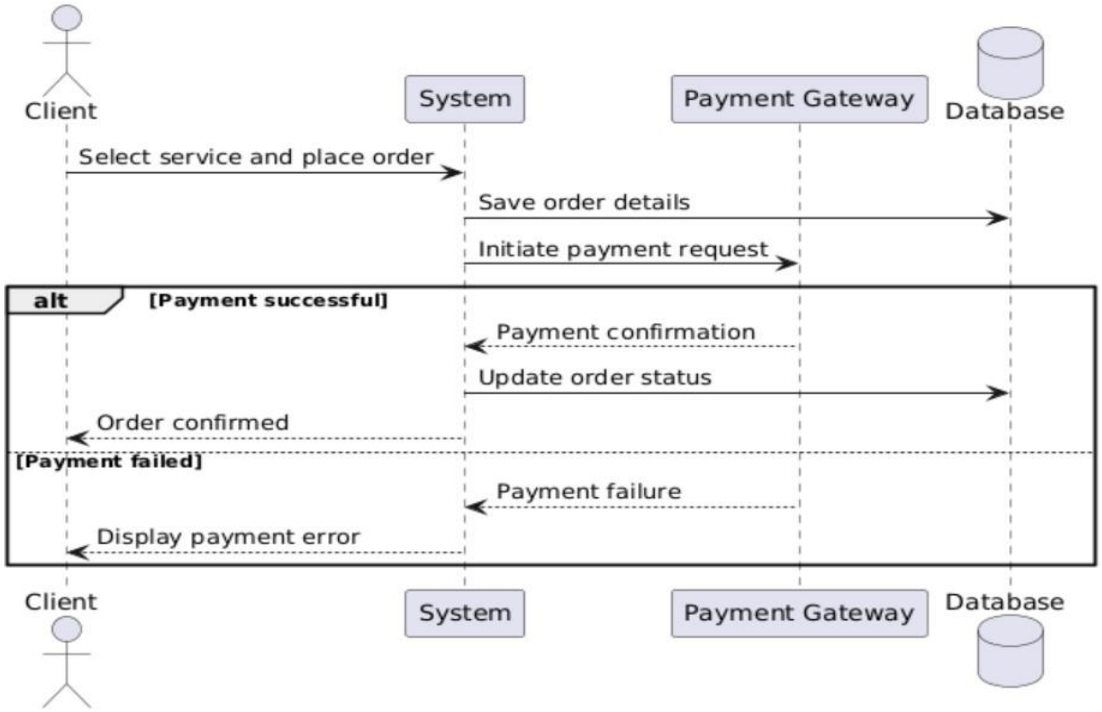


Figure 6: Place Service Order Sequence diagram

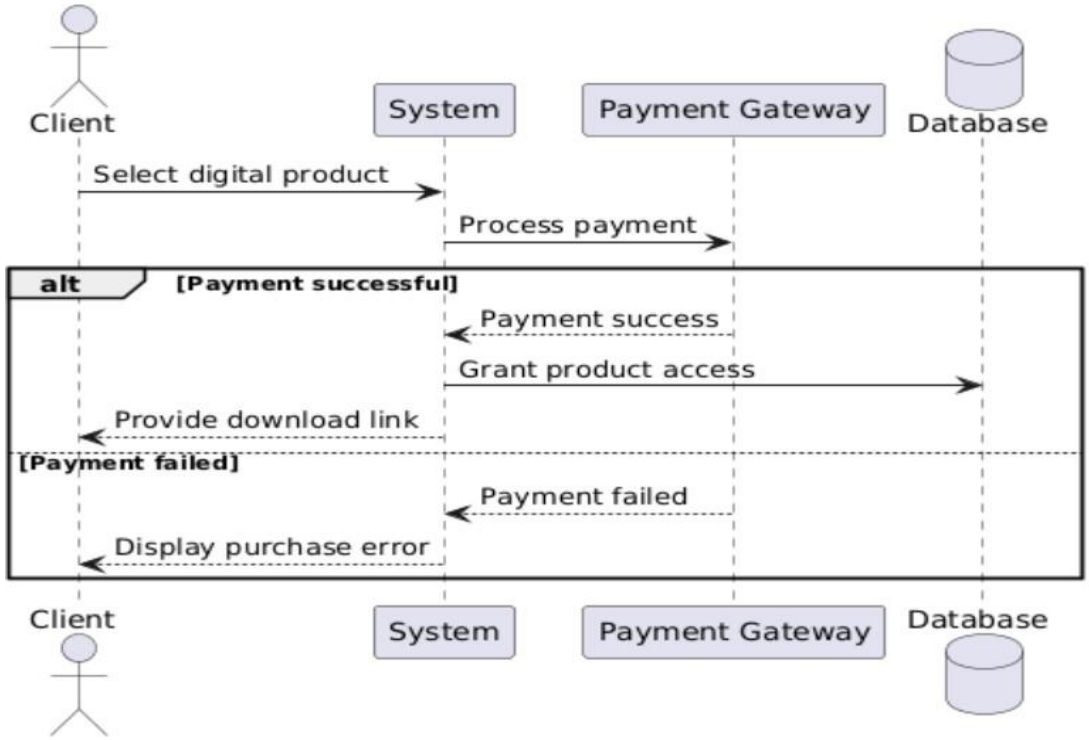


Figure 7: Purchase Digital Product Sequence diagram

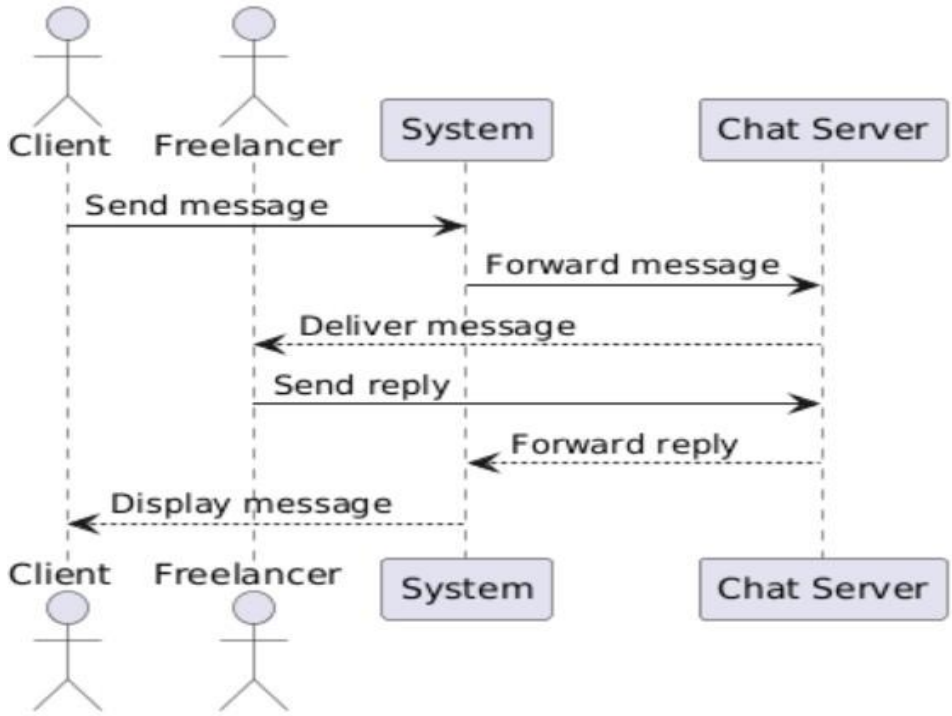


Figure 8: Real-Time Communication Product Sequence diagram

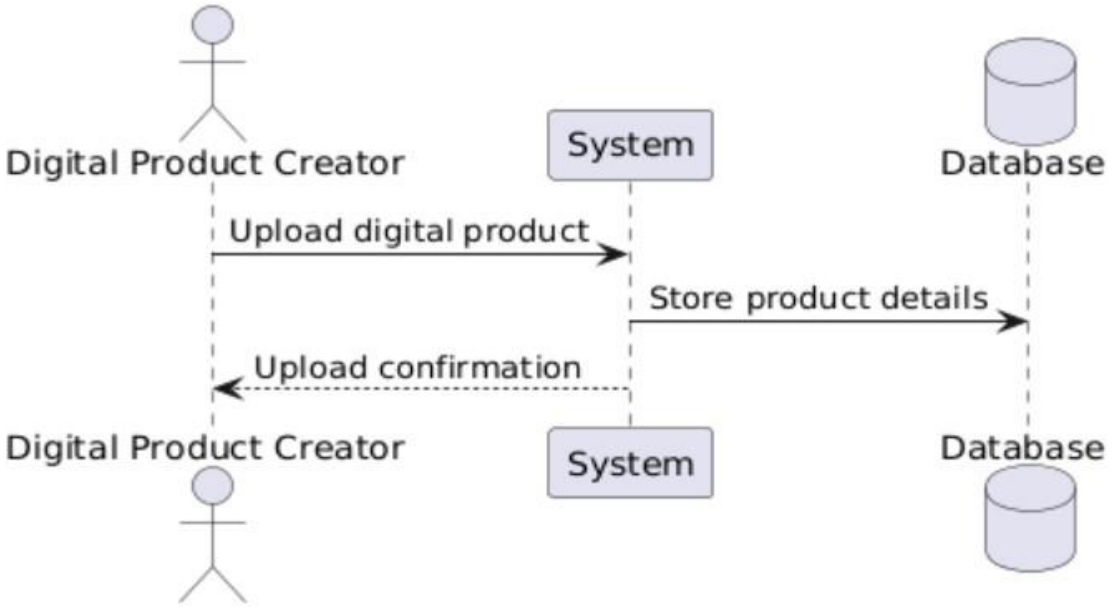


Figure 9: Upload Digital Product Sequence diagram

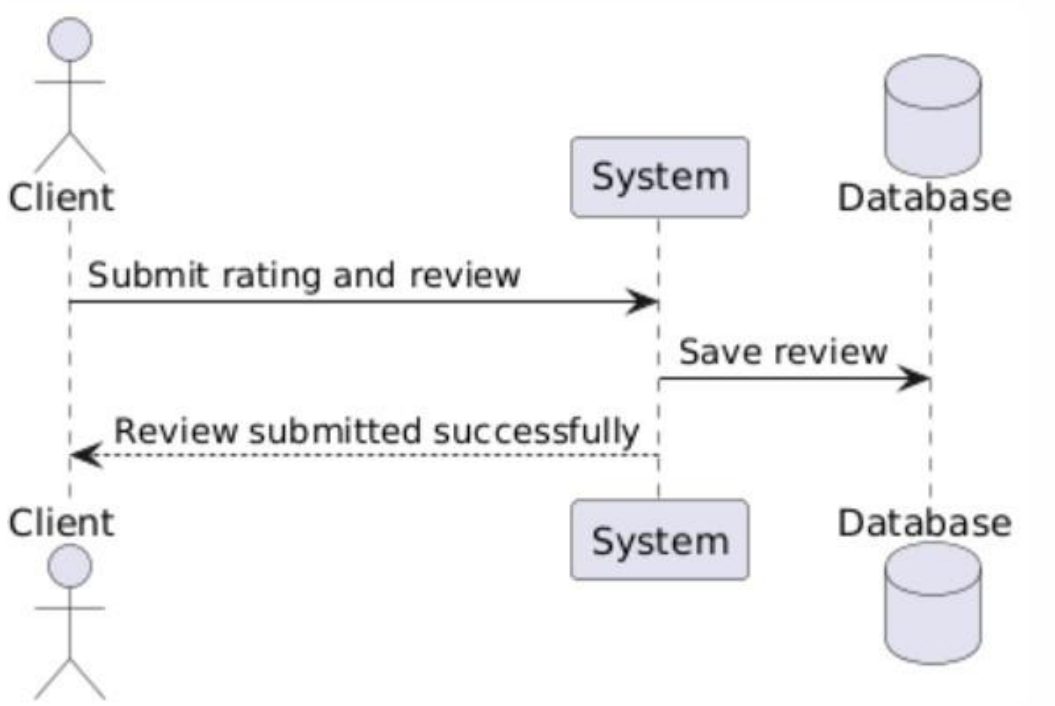


Figure 10: Rate and Review Sequence diagram

3.2.4 Activity Diagram

Activity diagrams illustrate the dynamic behavior of the system by modeling workflows and control flows. They show how different activities are coordinated to achieve a specific business process, including decision points, parallel actions, and termination states. The following activity diagrams represent the core operational processes of the Elite Ethiopia system.

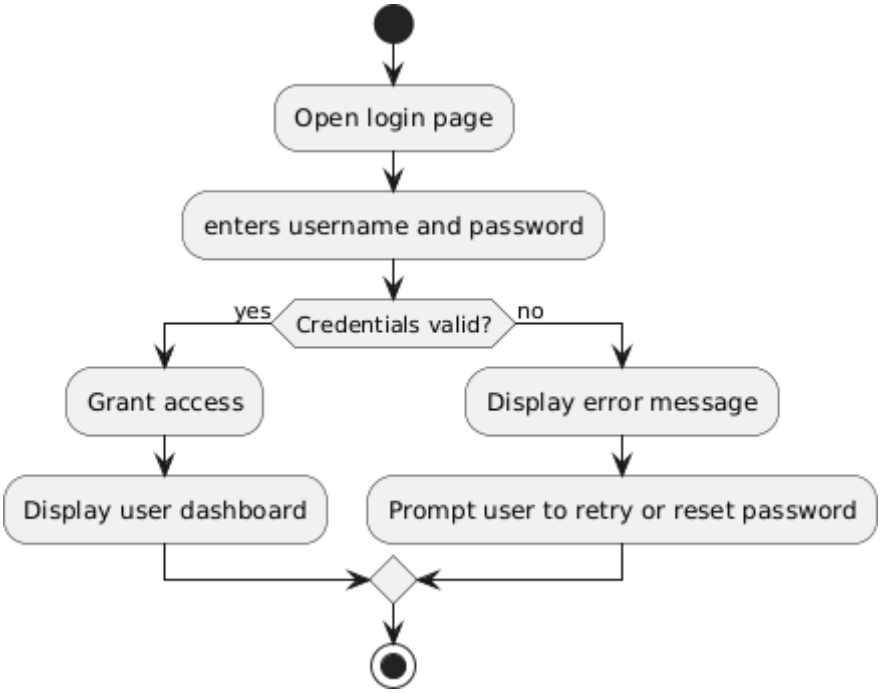


Figure 11: Login Activity diagram

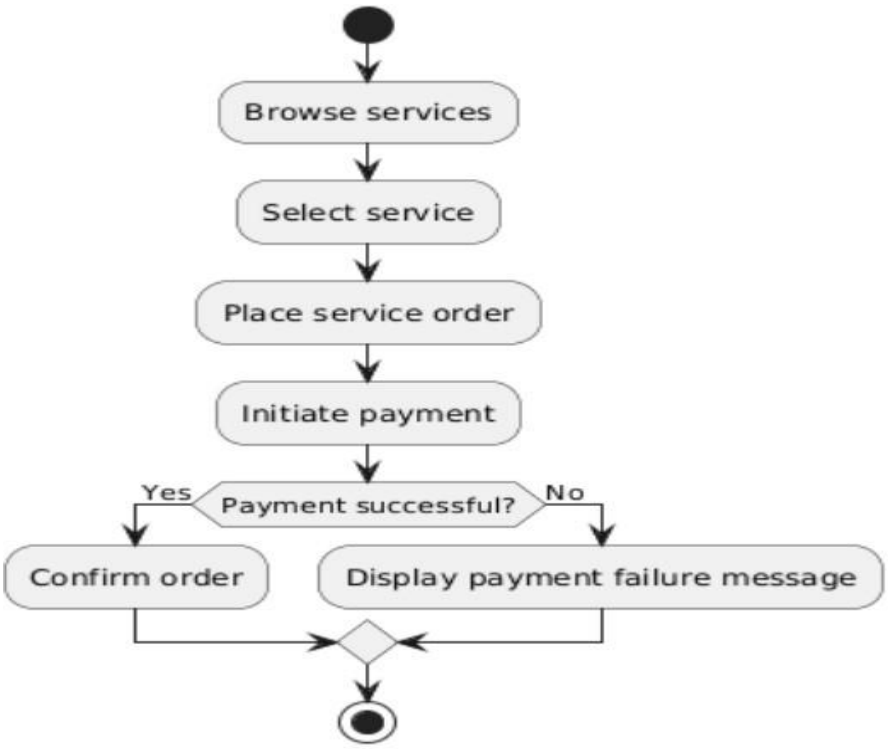


Figure 12: Placing a Service Order Activity diagram

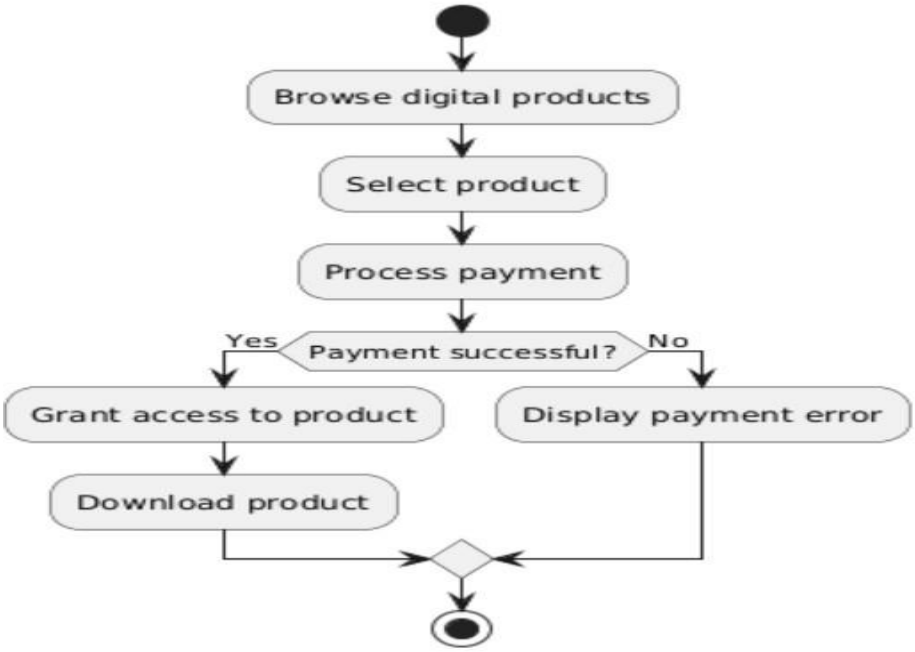


Figure 13: Purchasing Digital Products Activity diagram

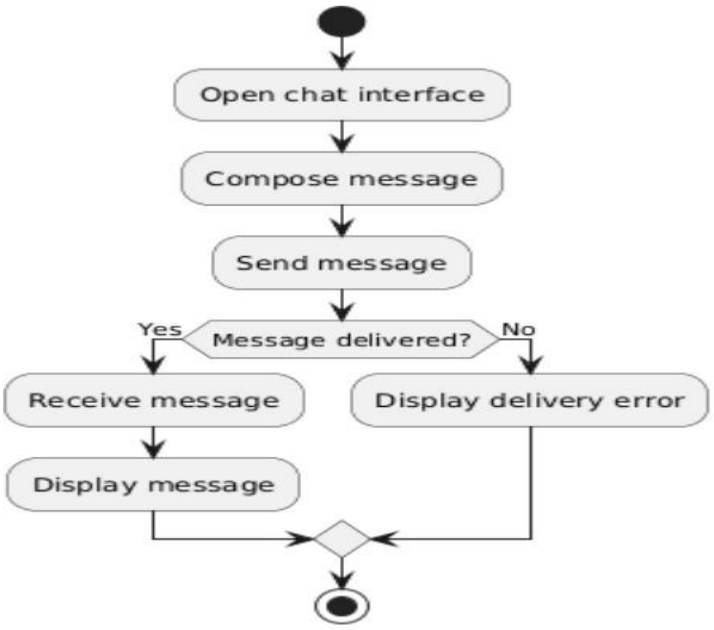


Figure 14: Real-Time Communication Activity diagram

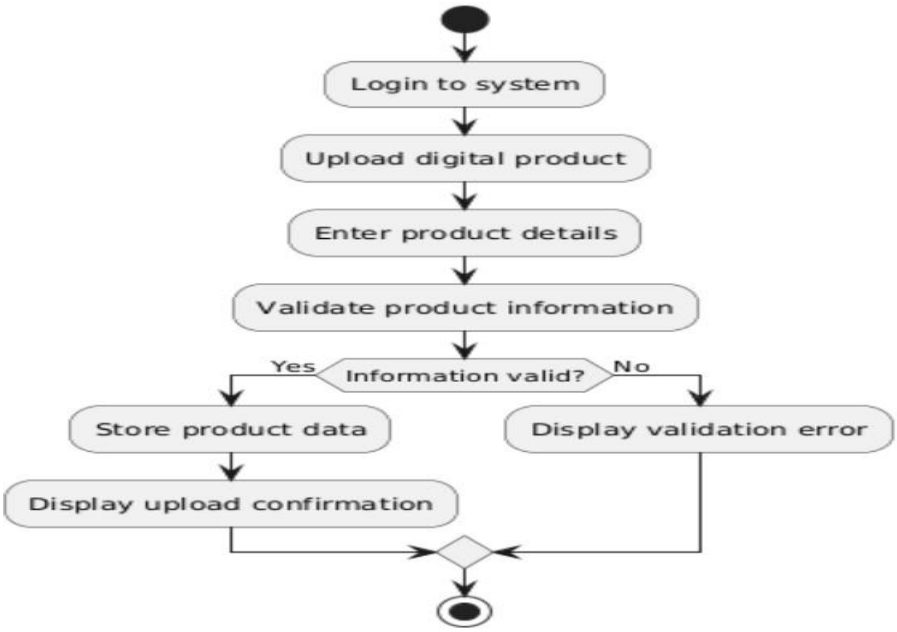


Figure 15: Uploading Digital Products Activity diagram

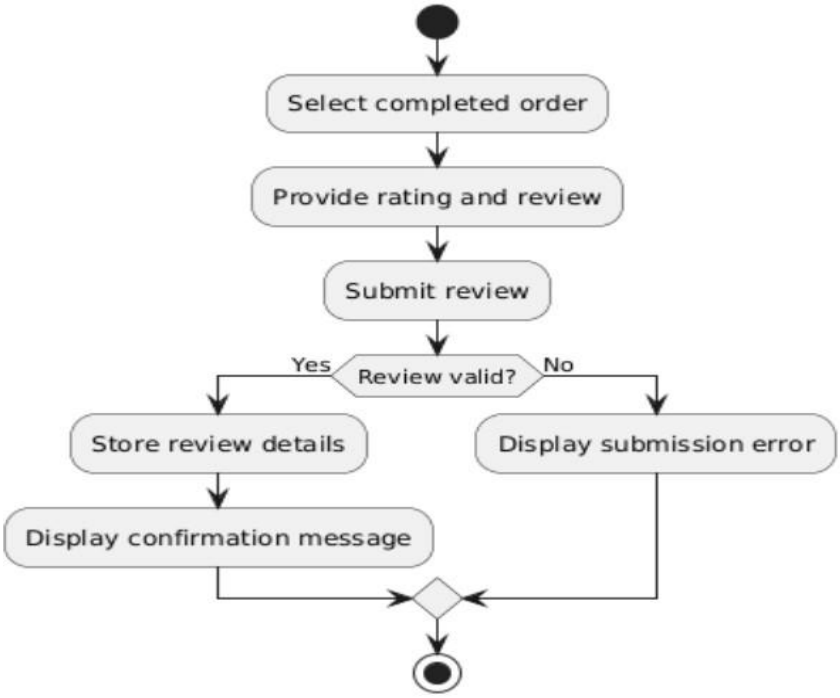


Figure 16: Rating and Reviewing Activity diagram

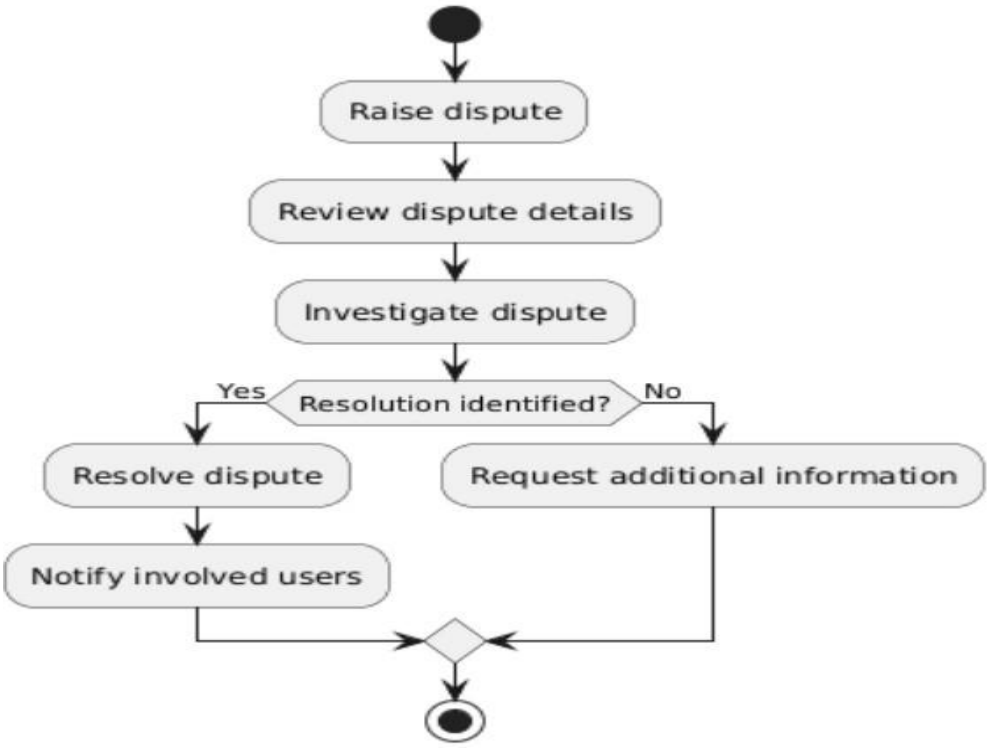


Figure 17: Dispute Management Activity diagram

3.2.5 Analysis level class diagram (conceptual modeling)

The analysis-level class diagram presents a high-level conceptual view of the Elite Ethiopia system. It identifies the main classes, their attributes, and the relationships among them without focusing on implementation details. This diagram helps in understanding the structure of the system and how different entities interact to support freelancing services, digital product transactions, communication, and administration.

The model focuses on **core business concepts**, user roles, and system objects that are essential for system functionality.

Main Classes Identified

- *User*

Represents all registered users of the system. It acts as a superclass for different user roles.

Key Attributes

- userId
- fullName
- email
- phoneNumber
- password
- role

- *Client*

Represents users who request services or purchase digital products.

Responsibilities

- Place service orders
- Purchase digital products
- Rate and review services

- *Freelancer*

Represents users who provide freelance services on the platform.

Responsibilities

- Create and manage service listings
- Communicate with clients
- Deliver services

➢ *DigitalProductCreator*

Represents users who upload and sell digital products.

Responsibilities

- Upload digital products
- Manage product listings

➢ *Service*

Represents freelance services offered on the platform.

Key Attributes

- serviceId
- title
- description
- price
- category

➢ *DigitalProduct*

Represents digital items sold on the platform.

Key Attributes

- productId
- title
- description
- price
- fileType

➢ *Order*

Represents a service order placed by a client.

Key Attributes

- orderId
- orderDate
- status
- totalAmount

➢ *Payment*

Represents payment transactions processed through local payment gateways.

Key Attributes

- paymentId
- amount
- paymentMethod
- paymentStatus

➢ *Review*

Represents ratings and feedback submitted by clients.

Key Attributes

- reviewId
- rating
- comment
- reviewDate

➢ *Chat*

Represents real-time communication between users.

Key Attributes

- chatId
- message
- timestamp

- *Dispute*

Represents issues raised regarding services or transactions.

Key Attributes

- disputeId
- reason
- status
- createdDate

- *Administrator*

Represents system administrators responsible for monitoring and control.

Responsibilities

- Manage users
- Resolve disputes
- Moderate services and products

Relationships Overview

- User is a **generalized class** for Client, Freelancer, DigitalProductCreator, and Administrator.
- A Freelancer **creates** one or more Services.
- A Client **places** Orders.
- An Order **includes** a Payment.
- A Client **writes** Reviews for completed Orders.
- A DigitalProductCreator **uploads** DigitalProducts.
- A Client **purchases** DigitalProducts.
- Users **communicate** through Chat.
- A Dispute is **raised by** a User and **handled by** an Administrator.

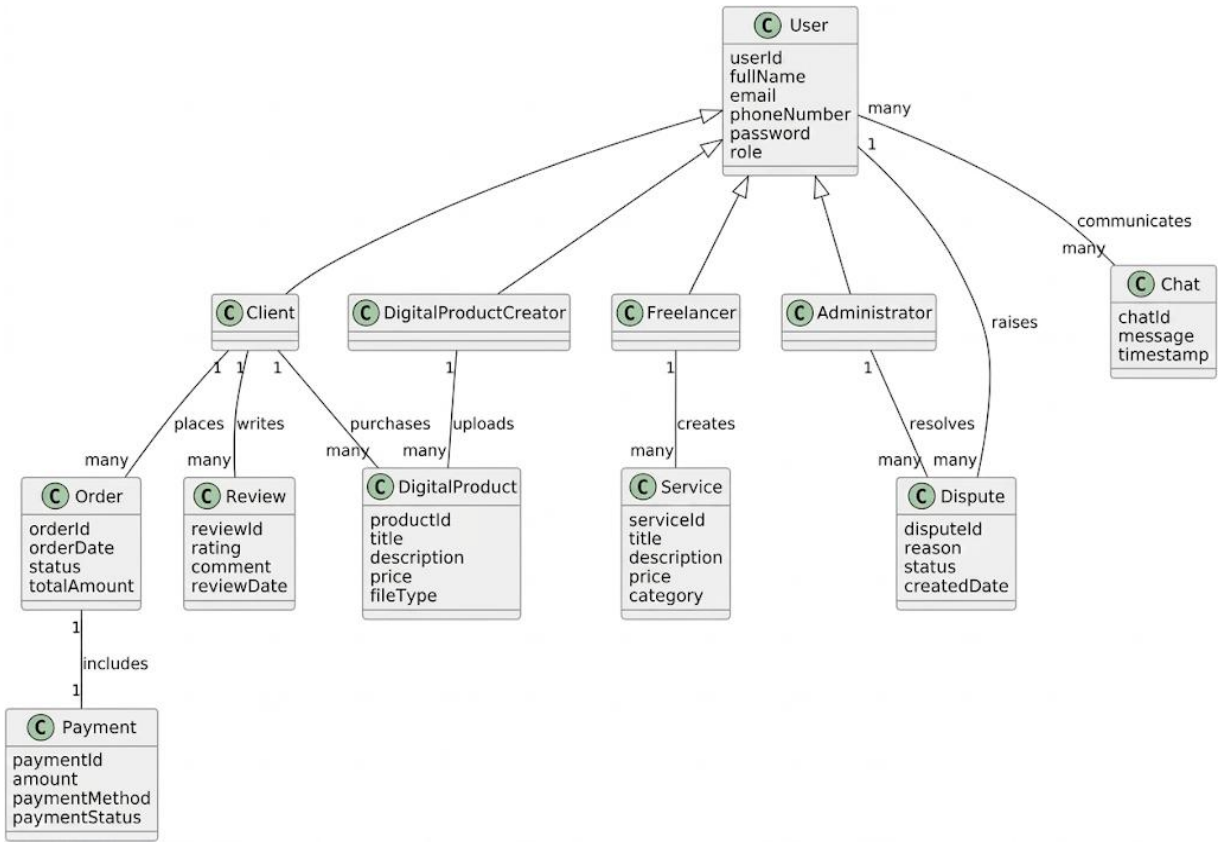


Figure 18: Analysis Level Class diagram

3.2.6 User Interface Prototyping

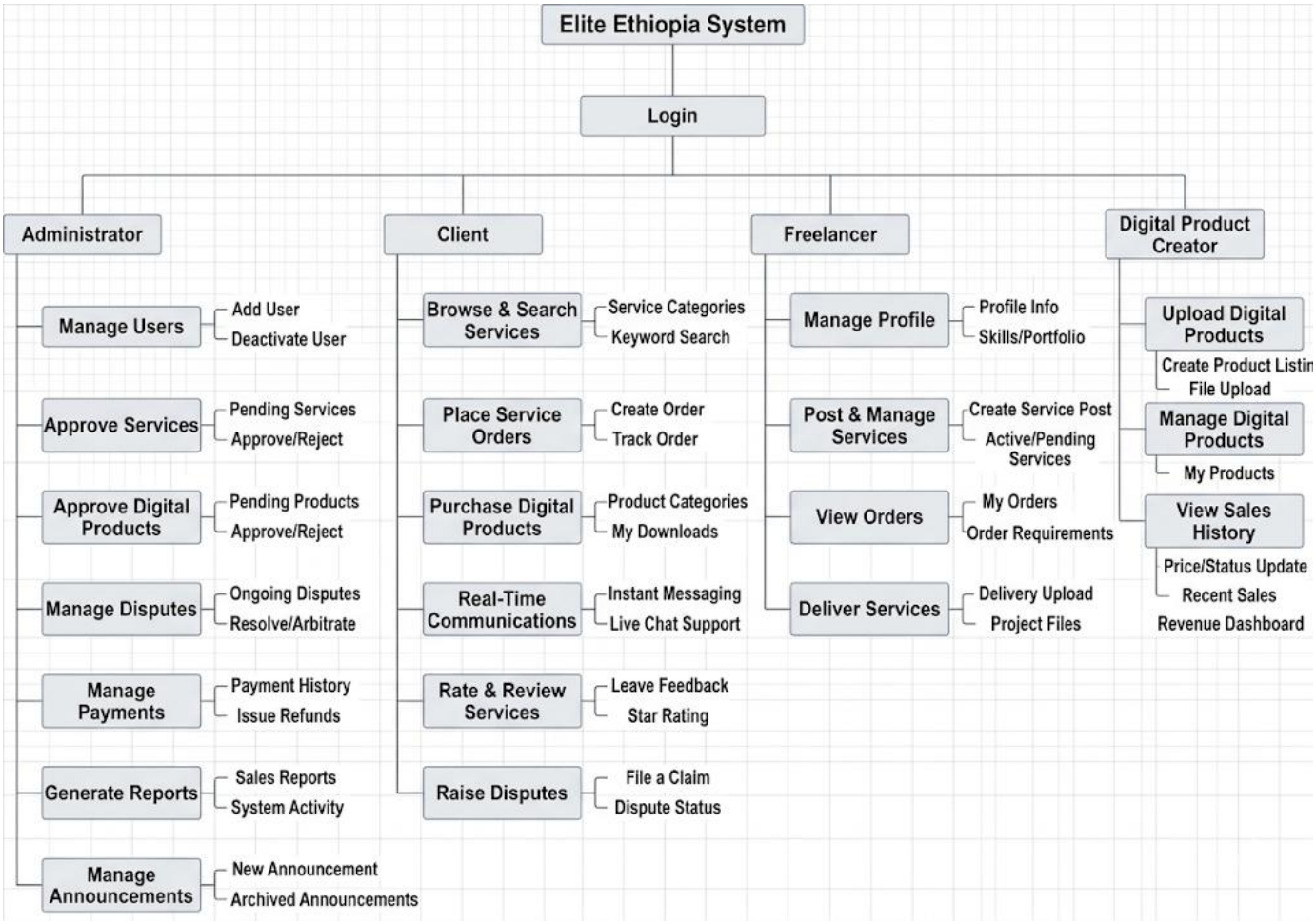


Figure 19: User Interface Prototyping

Chapter Four: System Design

4.1 Introduction

This chapter presents the **system design** of the **Elite Ethiopia System**. System design focuses on transforming the system requirements and analysis results into a structured and implementable solution. It defines how system components are organized, how they interact, and how responsibilities are distributed across different layers.

The Elite Ethiopia System is designed using a **layered class-type architecture** to ensure modularity, scalability, maintainability, and security. Each layer is assigned a specific responsibility and communicates with other layers through well-defined interfaces. This architectural approach minimizes dependency between components and supports future system enhancements.

4.2 Class Type Architecture

The Elite Ethiopia System follows a **multi-layered architectural model** that separates the system into logical layers based on functionality and responsibility. The main layers of the system include:

- User Interface Layer
- Controller / Process Layer
- Business / Domain Layer
- Persistence Layer
- System Layer

Each layer is described in detail below.

4.2.1 User Interface Layer

The **User Interface (UI) layer** is responsible for enabling interaction between users and the Elite Ethiopia System. It provides graphical interfaces through which users can access system functionalities.

This layer supports different user roles such as **Administrator, Client, Freelancer, and Digital Product Creator**. Each role is provided with role-specific interfaces that allow users to perform actions such as browsing services, placing orders, managing profiles, uploading digital products, processing payments, and handling disputes.

The UI layer handles user input and basic validation but does not contain business logic. All requests are forwarded to the Controller/Process layer for further processing, ensuring separation of concerns.

4.2.2 Controller / Process Layer

The **Controller/Process layer** acts as an intermediary between the User Interface layer and the Business/Domain layer. It manages the flow of requests and controls how system operations are executed.

This layer receives user requests from the UI layer, validates user sessions, checks access permissions, and invokes appropriate business services. It coordinates system activities such as service ordering, digital product purchasing, dispute submission, and real-time communication initiation.

By centralizing control logic, the Controller layer ensures consistent system behavior and simplifies system maintenance.

4.2.3 Business / Domain Layer

The **Business/Domain layer** contains the core business logic of the Elite Ethiopia System. It defines the rules, policies, and workflows that govern system operations.

This layer includes domain entities such as **User, Service, Order, Payment, DigitalProduct, Review, Dispute, Message, and Notification**. It enforces business rules such as payment validation, service completion status, rating eligibility, dispute resolution procedures, and access control rules.

The Business layer processes requests forwarded by the Controller layer and ensures all operations comply with defined business requirements before interacting with the Persistence layer.

4.2.4 Persistence Layer

The **Persistence layer** is responsible for data storage and retrieval. It abstracts the underlying database and provides a structured way to manage system data.

This layer handles operations such as saving and retrieving user information, service details, transaction records, reviews, dispute data, and communication logs. It ensures data consistency, integrity, and reliability through transactional mechanisms.

By isolating data access logic, the Persistence layer allows changes to the database technology without affecting higher-level system components.

4.2.5 System Layer

The **System layer** provides core infrastructure services that support the entire Elite Ethiopia System. These services include authentication, authorization, logging, error handling, configuration management, and security enforcement.

The System layer ensures secure access to system resources, manages user sessions, and monitors system activities. It also provides integration support for external services such as payment gateways and notification services.

This layer enhances system reliability, security, and performance by offering centralized system-level services.

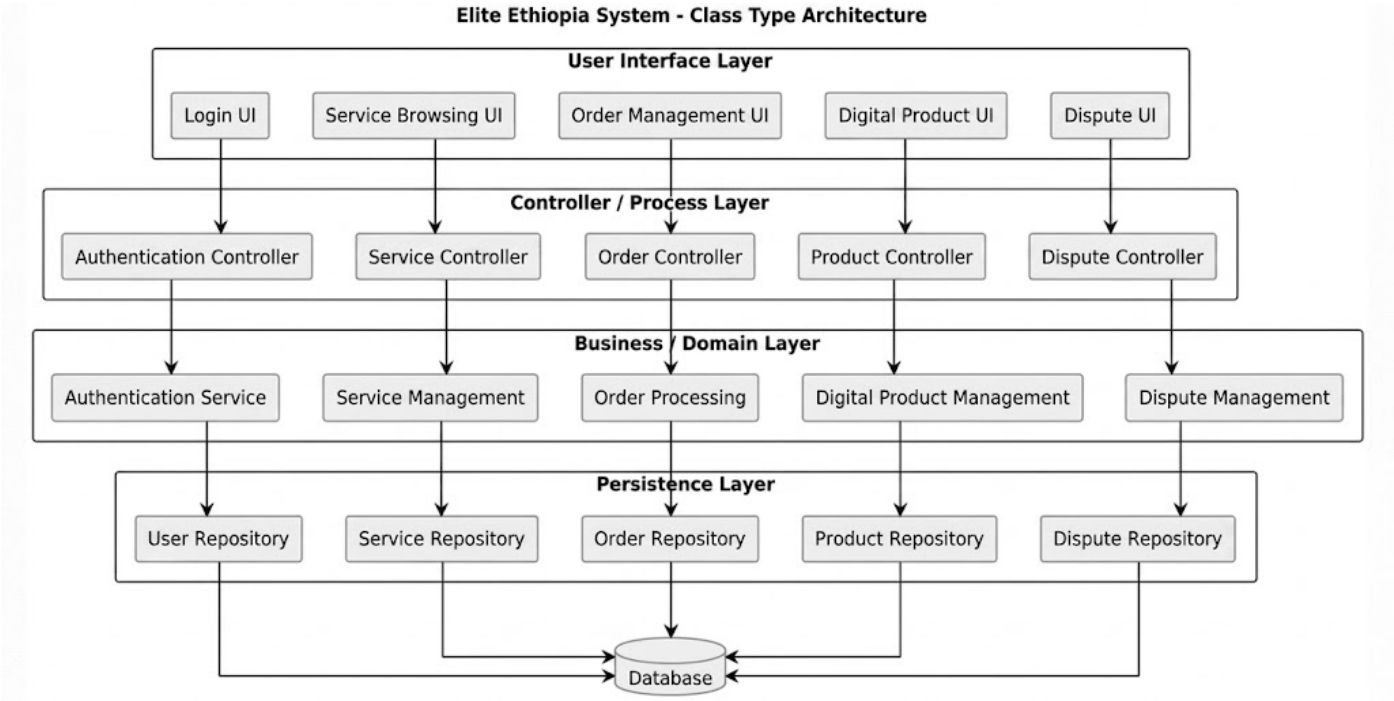


Figure 20: Class Type Architecture

4.3 Class modeling

Class modeling represents the **static structure** of the Elite Ethiopia system by identifying the main classes, their attributes, operations, and relationships.
This model focuses on how system entities such as users, services, orders, payments, digital products, and disputes interact with each other at the conceptual level.

The class diagram is used to:

- Define system data structures
- Show relationships between core entities
- Support implementation and database design

➢ Key Classes Identified

- User (Base class)
- Client

- Freelancer
- Digital Product Creator
- Administrator
- Service
- Service Order
- Digital Product
- Payment
- Review
- Dispute

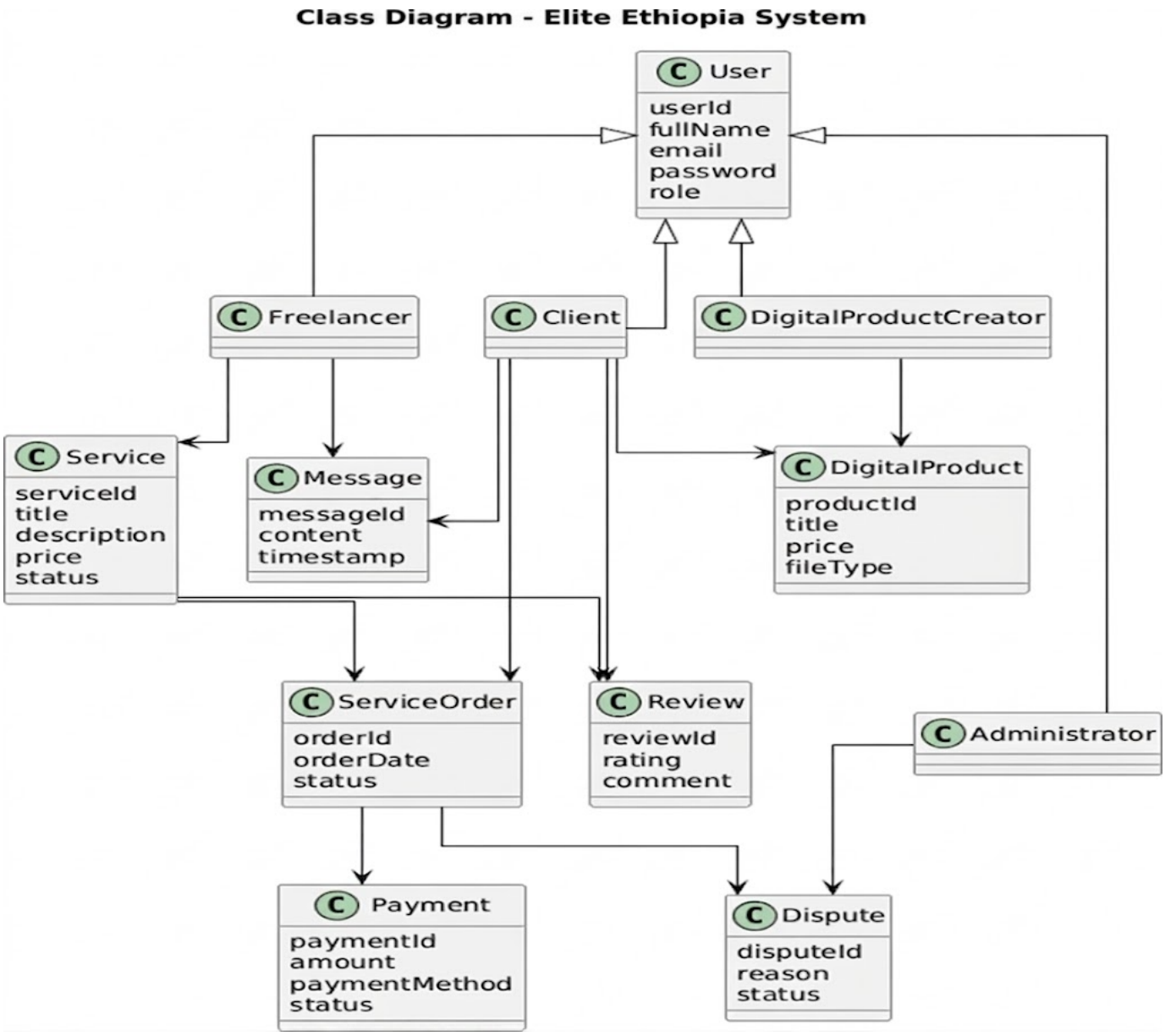


Figure 21: Class Diagram

4.4 State chart modeling

State chart modeling is used to represent the dynamic behavior of key components of the proposed Elite Ethiopia system by illustrating how they change from one state to another in response to specific events or actions. This modeling approach helps in understanding the system's operational flow over time, especially for processes that require multiple stages, validations, or approvals. In the Elite Ethiopia system, state charts are applied to model activities such as user registration, service request submission, request processing, approval or rejection, and service completion. Each state diagram demonstrates how user actions and system-generated events trigger transitions between different states, ensuring that defined business rules are followed and that service delivery processes remain consistent, and controlled.

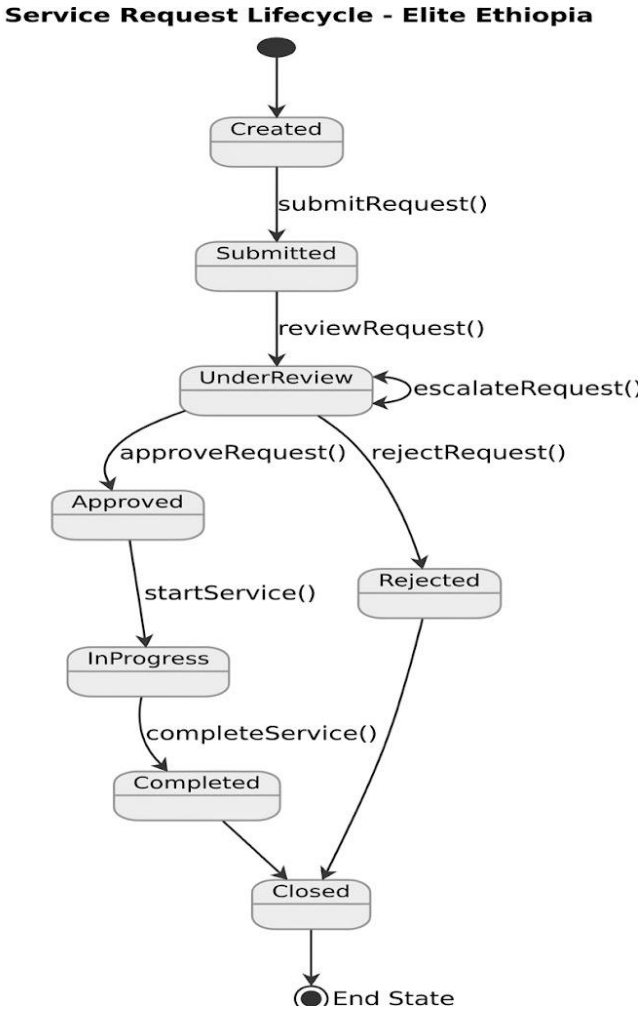


Figure 22: State Chart diagram for Service Request

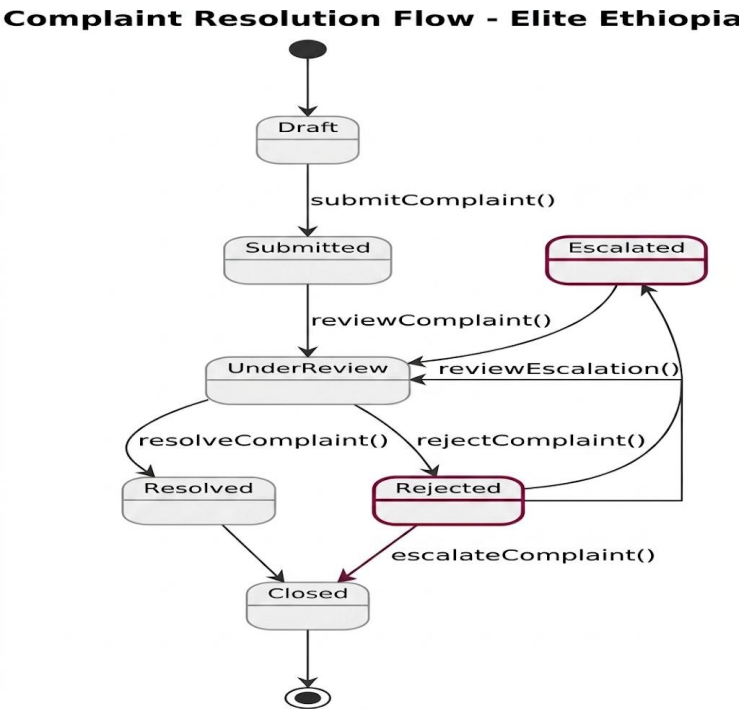


Figure 23: State Chart diagram Complaint Resolution

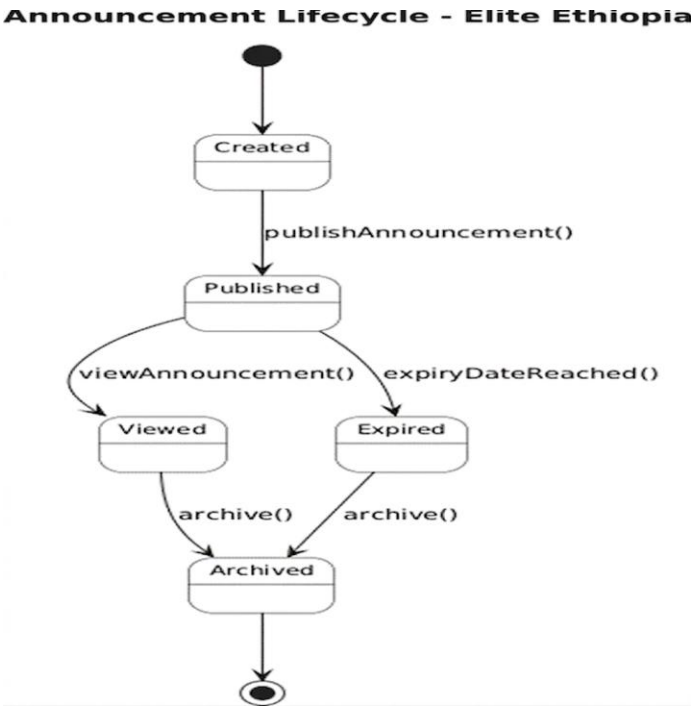


Figure 24:State Chart Diagram for Announcement Lifecycle

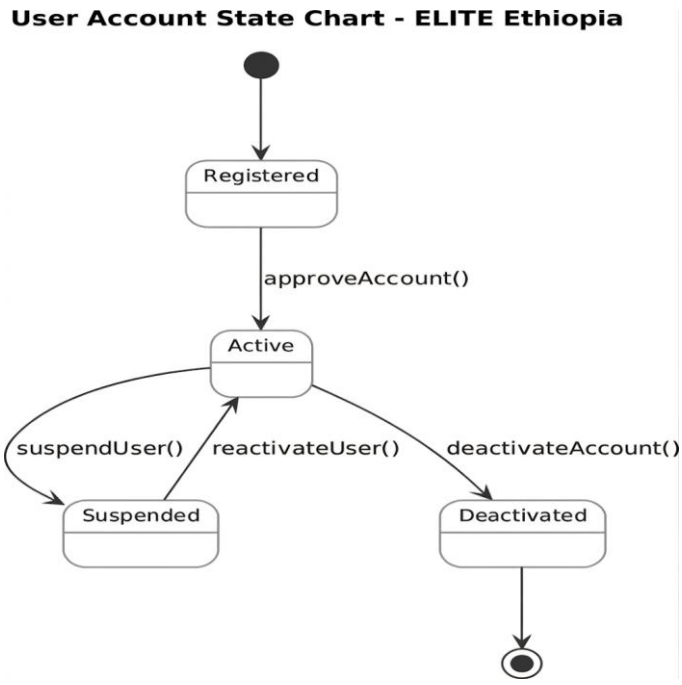


Figure 25:State Chart Diagram for User Account

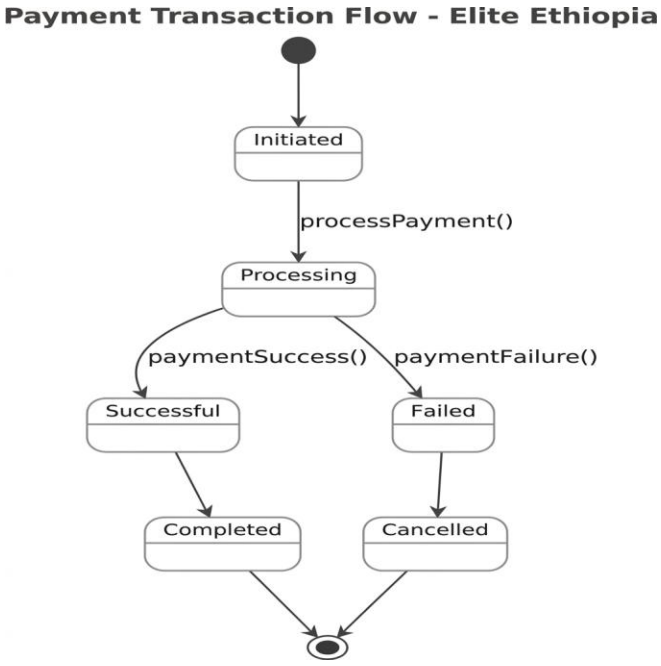


Figure 26: State Chart Diagram for Payment Transaction

4.5 Collaboration Modeling

Collaboration modeling illustrates how different system components and users within the proposed Elite Ethiopia system interact to complete specific tasks. Rather than emphasizing the sequence of events over time, collaboration diagrams focus on the structural relationships between system objects and the flow of messages exchanged among them. This approach helps to show how responsibilities are distributed across various system modules and how these modules work together to deliver electronic services.

In the Elite Ethiopia system, collaboration modeling is used to represent interactions involved in processes such as user registration, service request submission, request processing, approval or rejection of requests, and feedback delivery. These diagrams provide a clear understanding of user roles, communication links between system components, and the overall control flow required to ensure efficient and coordinated service delivery.

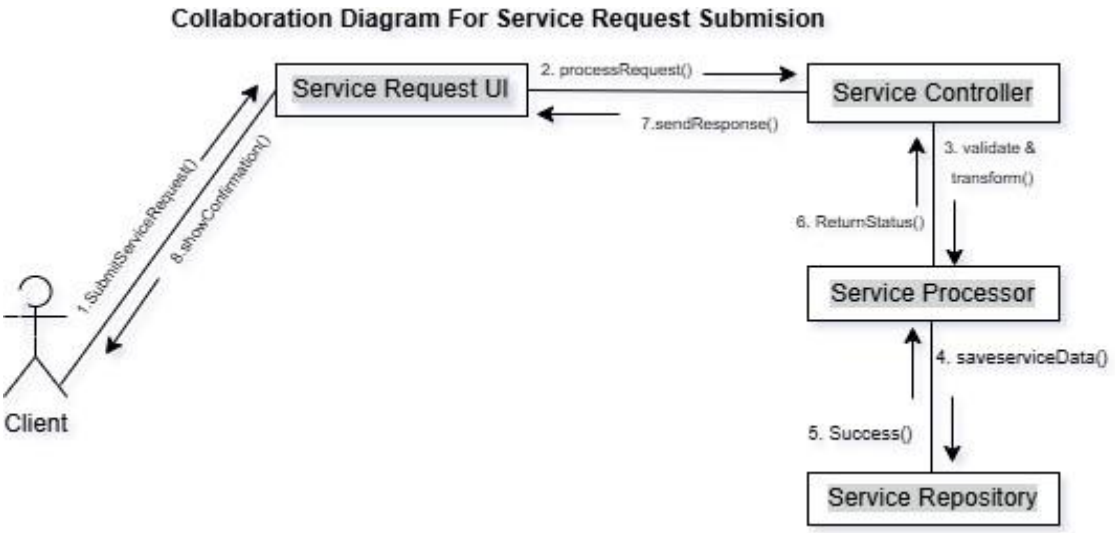


Figure 27: Collaboration Diagram for Service Request Submission

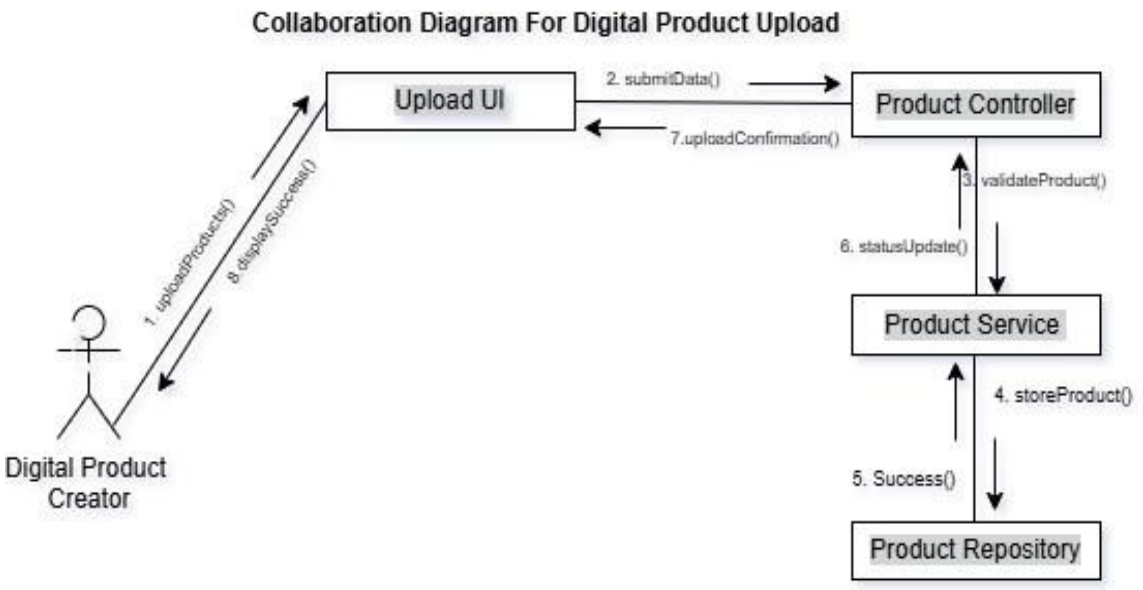


Figure 28: Collaboration Diagram for Digital Product Upload

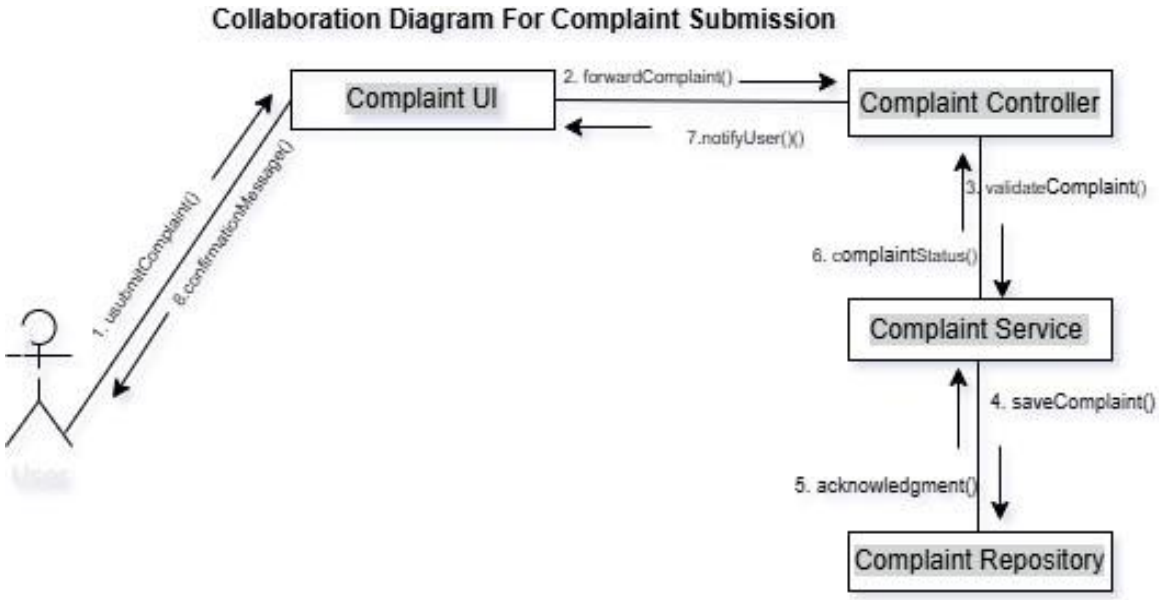


Figure 29: Collaboration Diagram for Complaint Submission

4.6 Component Modeling

Component modeling represents the Elite Ethiopia system as a set of modular, reusable, and independently manageable components, each designed to perform specific functions within the system. This modeling approach identifies logical groupings of related functionalities and illustrates how different components interact through well-defined interfaces.

In the proposed Elite Ethiopia system, component modeling is used to separate major modules such as the user interface, service request management, request processing, database management, and administrative control components. This separation enhances system maintainability, supports scalability, and improves the development process by allowing each component to be developed, tested, and updated independently without affecting the entire system.

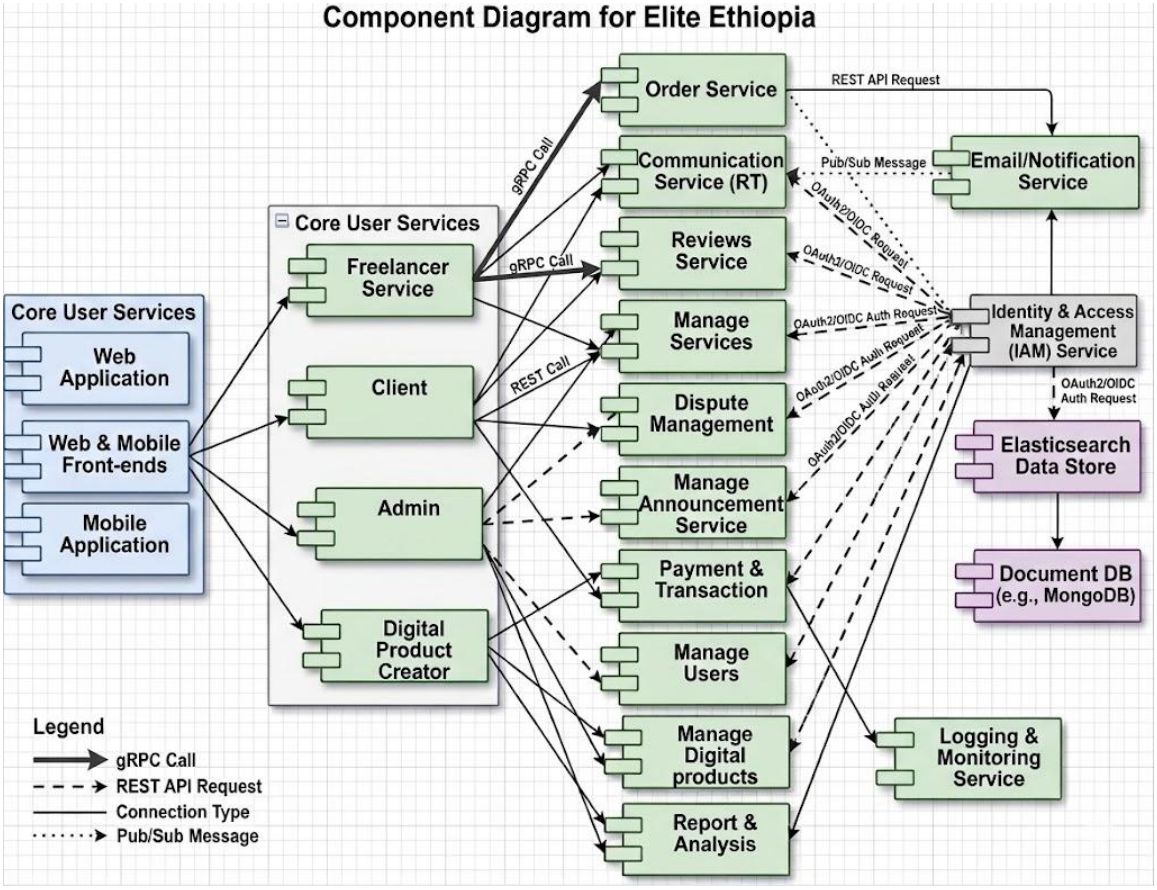


Figure 30: Component Diagram for Elite Ethiopia

4.7 Deployment modeling

Deployment modeling describes the physical architecture of the proposed Elite Ethiopia system by showing how software components are distributed across hardware resources. It illustrates the relationship between the system's logical components and the physical infrastructure, including client devices, web servers, application servers, databases, and supporting network services.

In the Elite Ethiopia system, deployment modeling explains how user requests originate from a web browser or client interface and are transmitted through the web server to the backend application and database for processing. This model helps clarify how system performance, availability, reliability, and security are achieved through proper allocation and configuration of hardware and software resources.

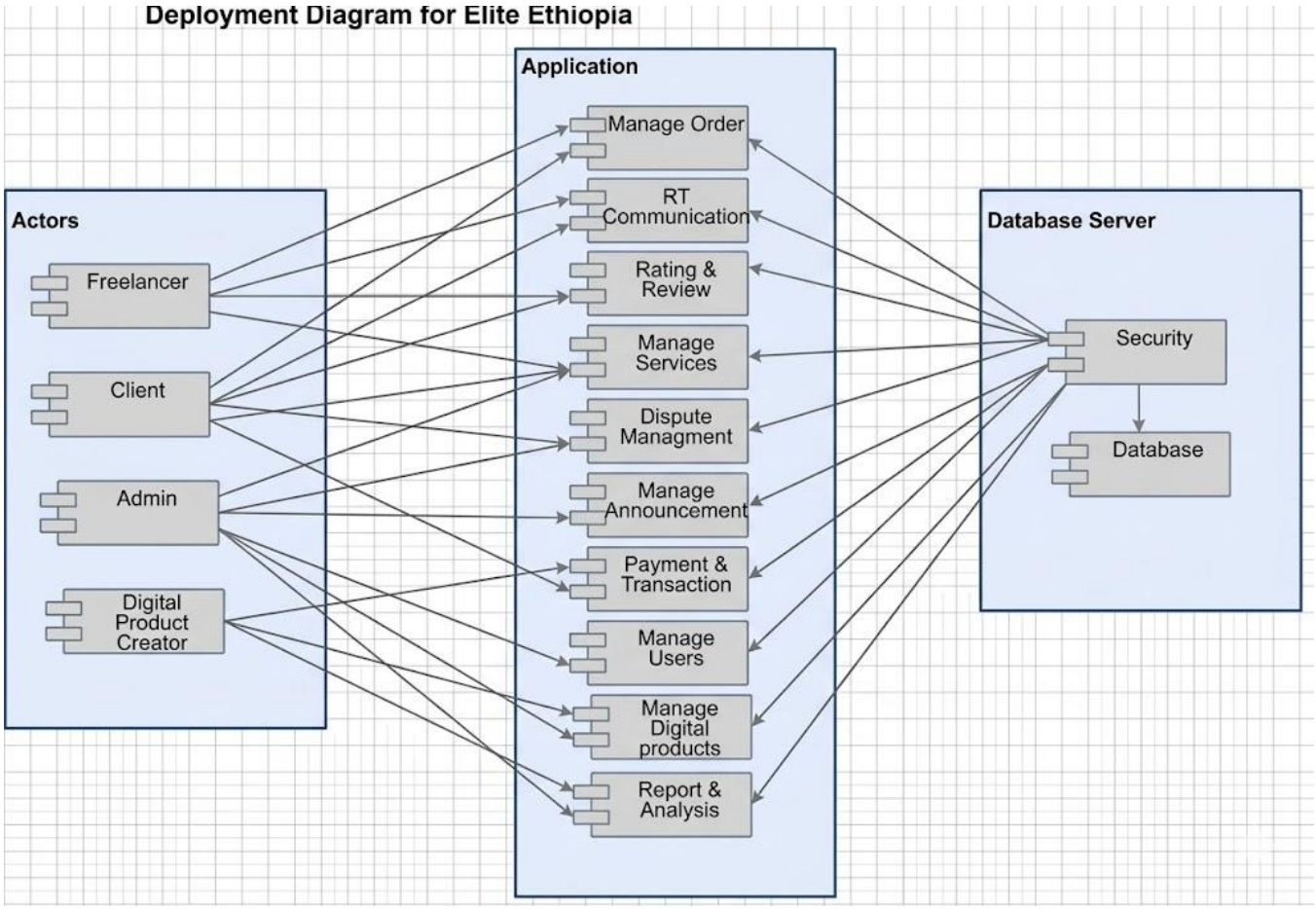


Figure 31: Deployment Diagram for Elite Ethiopia

4.8 Persistence modeling

Persistence modeling describes how data is stored, retrieved, and managed throughout the lifecycle of the Elite Ethiopia system. It defines the structure of the system's database, including the key entities, their attributes, and the relationships that exist between them.

In the proposed Elite Ethiopia system, persistence modeling ensures the accurate and consistent management of data such as user information, service requests, request statuses, administrative actions, and system records. This modeling approach supports data integrity, reduces redundancy, and enables efficient access to information required for effective electronic service delivery.

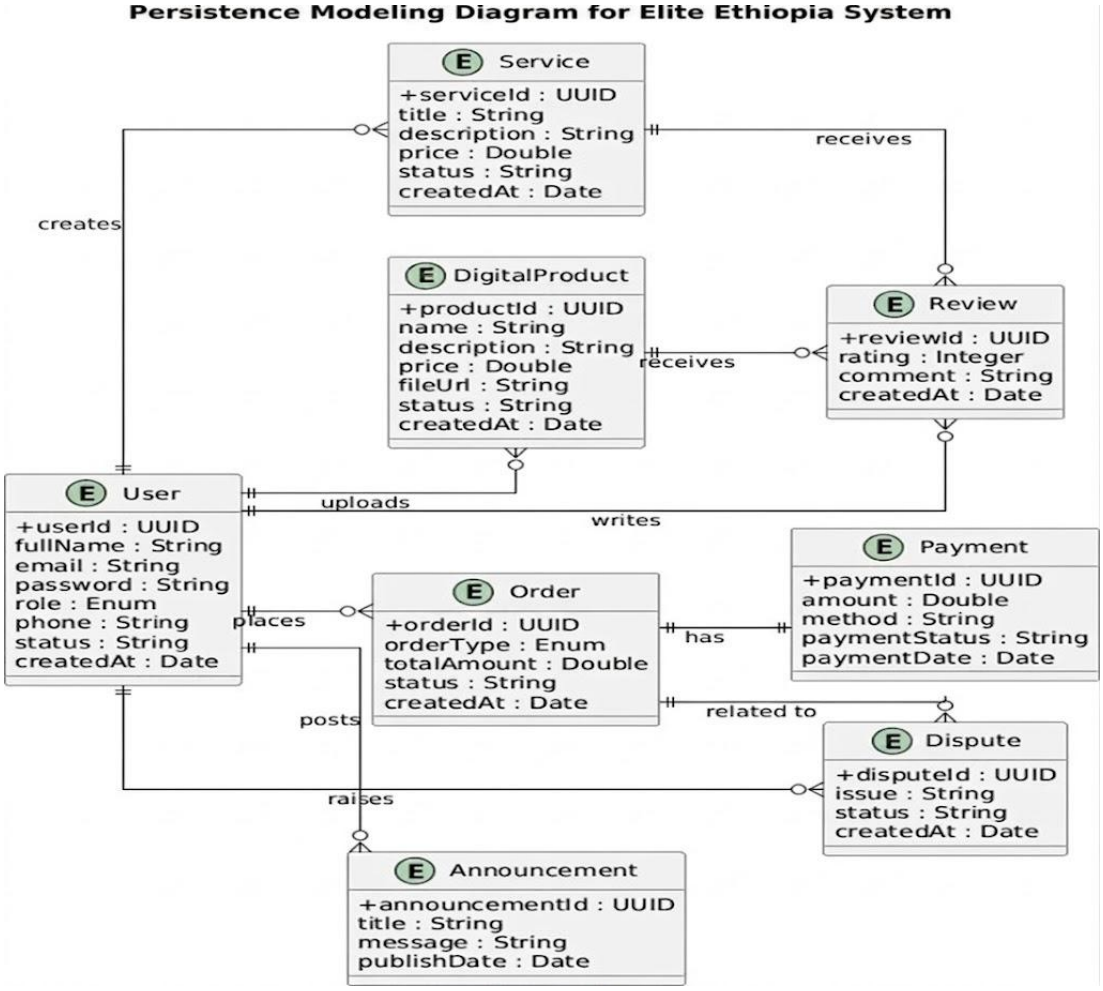


Figure 32: Persistence Diagram for Elite Ethiopia

4.9 User Interface design

User Interface (UI) Design focuses on developing an intuitive, accessible, and responsive front-end for all user roles within the proposed Elite Ethiopia system. The design approach emphasizes simplicity, consistency, and role-based access to ensure that users can efficiently interact with the system and complete their tasks without difficulty.

In the Elite Ethiopia system, interface components such as dashboards, input forms, service request pages, status tracking views, reports, and notification messages are designed according to user responsibilities. This role-specific design supports a smooth and logical workflow, minimizes user errors, enhances productivity, and encourages effective adoption of the electronic service platform.

Figure 33: UI of Login page

Figure 34: UI of Administrator Dashboard

Figure 35: UI of Client Dashboard

Figure 36: UI of Freelancer Dashboard

Figure 37: UI of Digital Product Creator Dashboard

Chapter Five: Implementation and Testing

5.1 Introduction

The implementation and testing stage represents the phase in which the proposed Elite Ethiopia system will be transformed from a conceptual and design-based solution into a fully functional system capable of supporting real users and service operations. At this stage, the system design models, database structures, and interface specifications developed in earlier chapters will be translated into executable software components and configured to operate within an appropriate technical environment.

This phase is not only on building the system but also on ensuring that all components function correctly when integrated. Attention will be given to verifying system functionality, stability, security, and ease of use. The objective is to ensure that the Elite Ethiopia system will reliably support activities such as service request submission, processing, tracking, and administrative management without errors or performance limitations.

5.2 Final Testing of the System

Final testing will be conducted to validate that the complete Elite Ethiopia system functions as intended and meets all predefined functional, technical, and usability requirements. This phase will ensure that all system modules operate correctly both independently and when integrated, under realistic operating conditions. The testing process will also confirm that users can successfully access services, submit requests, track service status, and receive feedback without errors.

The following testing approaches will be employed:

- **Unit Testing:** Each module of the Elite Ethiopia system will be tested individually to confirm that it performs its intended function correctly. Any errors identified at this stage will be corrected without affecting other modules.
- **Integration Testing:** All system modules will be combined and tested to verify proper interaction, data exchange, and compatibility among components. Issues identified during integration will be resolved before proceeding further.

- **User Interface Testing:** Interface elements such as forms, buttons, input fields, validation messages, and navigation controls will be tested to ensure correctness, responsiveness, and user friendliness.
- **Usability Testing:** This testing will evaluate how easily users can learn and operate the system, prepare inputs, interpret outputs, and complete service-related tasks efficiently.

Successful completion of final testing will confirm that the Elite Ethiopia system is stable and ready for deployment.

5.3 Hardware and Software Acquisition

The Elite Ethiopia system will be developed as a web-based application and requires no additional hardware beyond existing infrastructure. Development and testing activities were carried out using standard computer systems with adequate processing power, memory, and storage capacity. For deployment, the system will be hosted on a server or cloud-based environment capable of supporting web applications and database services.

Required software resources will include an operating system, web server, programming environment, database management system, and web browsers. Open-source tools and frameworks will be utilized to minimize cost and ensure flexibility, scalability, and ease of maintenance. These resources will provide a reliable platform for hosting and managing the Elite Ethiopia system.

5.4 User Manual Preparation

User documentation will be prepared to guide users on how to effectively interact with the Elite Ethiopia system. The user manual will provide clear and concise instructions on essential system operations such as user registration, login procedures, service request submission, request tracking, and logout processes.

Separate guidelines will also be prepared for system administrators, detailing tasks such as user management, service request processing, system monitoring, and data maintenance. The availability of these manuals will help reduce user errors, improve usability, and enhance overall system acceptance.

5.5 Training

Training sessions will be organized to ensure that users and administrators are able to use the Elite Ethiopia system effectively. The training will focus on demonstrating system features, explaining operational workflows, and addressing common challenges that may arise during usage.

End users will be trained on how to access services electronically, submit service requests, and monitor service progress, while administrators will receive training on managing users, processing requests, and maintaining system records. These training activities will help improve user confidence and ensure smooth adoption of the system.

5.6 Installation Process

The installation process will involve deploying the Elite Ethiopia system on the designated server or hosting environment. This will include installing and configuring the required software components, setting up the database, and uploading the system application files.

After installation, configuration settings such as database connections, security parameters, and access controls will be verified. The system will then be tested in the live environment to ensure proper functionality, stability, and accessibility before full deployment.

5.7 Start-up Strategy

A phased start-up strategy will be adopted to introduce the Elite Ethiopia system into operational use gradually. Initial deployment will be limited to a controlled group of users in order to monitor system performance and identify any unforeseen issues.

User feedback will be collected during the initial usage period and used to improve system functionality and usability. Ongoing support and maintenance procedures will be established to ensure continuous system availability, reliability, and performance.

Chapter Six: Conclusions and Recommendations

6.1 Conclusions

The proposed Elite Ethiopia system represents a significant improvement in the way services are delivered and managed within the organization. It is designed to address common challenges associated with manual service processes, such as delays, lack of transparency, poor record keeping, and limited user feedback. By introducing a centralized and web-based platform, the system aims to streamline service delivery, reduce administrative workload, and improve overall efficiency.

Through the Elite Ethiopia system, users will be able to submit service requests electronically, track the progress of their requests, and receive timely responses without the need for repeated physical visits. This approach is expected to save time for both users and service providers while ensuring that requests are handled in a fair and consistent manner. Administrators will also benefit from improved control over service workflows, easier access to records, and better decision-making through organized and reliable data.

Beyond technical improvements, the Elite Ethiopia system is expected to promote transparency, accountability, and trust between users and service providers. By replacing fragmented and manual processes with a structured digital solution, the system supports a more responsive and user-centered service environment. Overall, the proposed Elite Ethiopia system is intended to enhance service quality, improve user satisfaction, and support efficient organizational operations.

6.2 Recommendations

To ensure the effective adoption and long-term success of the proposed Elite Ethiopia system, several recommendations are made. First, it is important that users and administrators receive adequate training before and after deployment. Regular training sessions, supported by simple user guides and help resources, will help users feel confident and comfortable using the system.

Continuous system maintenance and improvement should also be prioritized. Establishing a dedicated support or technical team to monitor system performance, fix issues, and implement updates based on user feedback will ensure that the platform remains reliable and relevant over

time. As the system becomes stable, it is recommended that additional services and departments be gradually integrated to maximize its benefits.

Data security and privacy should be treated as a top priority. Regular security reviews, data backups, and access control measures should be enforced to protect sensitive information and comply with organizational policies. Creating an open feedback channel where users can report issues or suggest improvements will further enhance system usability and responsiveness.

In the future, the Elite Ethiopia system may be integrated with other existing platforms such as payment systems, records management systems, or mobile applications to further simplify service delivery and reduce duplicated efforts. Periodic evaluation of service workflows is also recommended to ensure that processes remain efficient, fair, and aligned with organizational goals. By implementing these recommendations, the organization can ensure that the Elite Ethiopia system continues to deliver value and improve service delivery for all stakeholders.

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